

University of Pisa
Master Thesis
A.Y 2026-2027

From Patents to Places: A Spatial-Causal Framework for Linking Environmental Risk Indicators to Technological Failure Modes

Thesis supervisor:
Prof. Gualtiero Fantoni
Profssa. Irene Spada

Submitted by:
Rafael Urbina

October 28, 2026

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Key Terms

Term 1 — Short definition of the term.
Term 2 — Short definition of the term.
Term 3 — Short definition of the term.

Contents

1 Introduction **5**
- 1.1 Context and motivation . . . 5
- 1.2 Problem statement . . . 6
  - 1.2.1 The Structural and Physical Data Deficit . . . 6
  - 1.2.2 Scalability Barriers in Traditional Diagnostics . . . 7
  - 1.2.3 Data Isolation, Fragmentation, and Semantic Heterogeneity . . . 7
  - 1.2.4 The Gap in Integrated Decision Frameworks . . . 7
- 1.3 Research questions and objectives . . . 7
  - 1.3.1 Main Research Question . . . 8
  - 1.3.2 Secondary Research Objectives . . . 8
- 1.4 Structure of the thesis . . . 9

2 Literature review & related work **11**
- 2.1 Literature review . . . 11
  - 2.1.1 The Foundations, Data Fusion and Predictive Maintenance (2005–2017) . . . 11
  - 2.1.2 Cyber-Physical Systems and Intelligent Water Telemetry (2018–2020) . . . 12
  - 2.1.3 Semantic Web Integration, Structural Monitoring, and Patent Intelligence (2021–2023) . . . 13
  - 2.1.4 Scalable Remote Sensing, Physics-Informed ML, and Closed-Loop Control (2024–2026) . . . 15
- 2.2 State of the art . . . 17
  - 2.2.1 Non-Invasive Subterranean Pipeline Mapping and Leak Detection . . . 17
  - 2.2.2 Bidirectional Telemetry, Edge Computing, and Digital Water Software . . . 17
  - 2.2.3 Integrated Decision Support and Closed-Loop Control Frameworks . . . 18
  - 2.2.4 Application context . . . 18
- 2.3 Research gaps . . . 22

3 Data Sources, Extraction, & Transformation **23**
- 3.1 Data Source Discovery . . . 23
  - 3.1.1 Exploration of Repositories and Portals . . . 23
  - 3.1.2 Automated Metadata Harvesting and API Discovery . . . 23
- 3.2 Data Selection . . . 28
- 3.3 Data Transformation and cleaning . . . 31
  - 3.3.1 Geometry-Specific Transformation Principles . . . 31
  - 3.3.2 Pipeline Integration Methodology and Execution Architecture . . . 32
- 3.4 Data summary statistics . . . 33
  - 3.4.1 Ordinal and Discrete Quantitative Metrics . . . 33
  - 3.4.2 Categorical Attributes and Completeness Analysis . . . 34
  - 3.4.3 Spatial statistics: mathematical framework . . . 35
  - 3.4.4 Spatial Comparative Results and Analysis . . . 35
- 3.5 Patent data case . . . 43
  - 3.5.1 Source selection . . . 43
  - 3.5.2 Query design . . . 43
  - 3.5.3 Patent processing & summary statistics . . . 45
- 3.6 Ontology and key term lists . . . 47
  - 3.6.1 Phase 1: Terminology Extraction and Domain Lexicon Mining . . . 47
  - 3.6.2 Phase 2: Programmatic Knowledge Graph Generation and RDF Encoding 48

4 Methodology **50**
4.1 Overview . . . 50
4.2 Conceptual Pipeline . . . 50
4.3 Data Flow . . . 51
4.4 Design Considerations . . . 53
4.5 Data Collection and Preparation . . . 53
4.5.1 Geospatial Data Processing . . . 53
4.5.2 Patent Data Processing . . . 53
4.6 Spatial Data Analysis . . . 54
4.6.1 Spatial Correlation Analysis . . . 54
4.7 Patent Data Analysis . . . 54
4.7.1 Causal Relation Extraction . . . 54
4.7.2 Ontology Integration . . . 54
4.8 Integration Process . . . 55
4.8.1 Spatial-Patent Linking . . . 55
4.8.2 Final Integration . . . 55
4.9 Pipeline Architecture . . . 55
4.9.1 Data Ingestion . . . 55
4.9.2 Processing Workflow . . . 55
4.10 Output and Applications . . . 55
4.10.1 Spatial Triggers . . . 55
4.10.2 Patent-Driven Insights . . . 55
4.10.3 Integrated Results . . . 56

5 Implementation and Experimental Setup **56**
5.1 Implementation Overview . . . 56
5.2 Implementation Details . . . 56
5.3 Experimental Setup . . . 57
5.4 Evaluation Protocol . . . 57
5.5 Experimental Scenarios . . . 57
5.6 Evaluation Metrics . . . 58

6 Results and Analysis **58**
6.1 Overview . . . 58
6.2 Empirical Findings . . . 59
6.2.1 Data Characteristics . . . 59
6.2.2 Performance Metrics . . . 59
6.2.3 Comparison with State-of-the-Art . . . 60
6.2.4 Statistical Significance . . . 60
6.3 Business & Technical Interpretation . . . 61
6.3.1 Technical Interpretation . . . 61
6.3.2 Business Interpretation . . . 61
6.4 Model Explainability & Limitations . . . 62
6.4.1 Model Explainability . . . 62
6.4.2 Limitations . . . 63
6.5 Scenario Analysis: A Business Case Study . . . 63
6.6 Summary of Key Results . . . 63

7 Conclusion and Future Work **64**
7.1 Summary of Contributions 64
7.2 Final Business and Technical Takeaways 64
7.3 Recommendations for Future Work 65

A Appendices **74**
A.1 Supplementary Data 74
A.2 Additional Results 74
A.3 Technical Specifications 74

1 Introduction

1.1 Context and motivation

Water scarcity has emerged as one of the most critical structural threats of the twenty-first century, transcending local ecological concerns to become a defining barrier to global economic and social stability [1]. According to the United Nations University (UNU-INWEH) and the World Economic Forum, the biosphere has transitioned from a state of transient water crises into a persistent era of *global water bankruptcy* [2]. This condition of water bankruptcy represents a systemic, often irreversible depletion of natural hydrological capital, wherein human water extraction and pollution consistently outpace renewable aquifer recharge and riverine flows [2]. Currently, over 2.2 billion people globally live without access to safely managed drinking water, 3.5 billion lack basic sanitation, and nearly 4 billion experience severe seasonal water scarcity for at least one month per year [2].

A primary driver of this global hydrological insolvency is the severe physical and administrative inefficiency within municipal Water Distribution Networks (WDNs). Globally, the volume of Non-Revenue Water (NRW)—treated water that is lost through pipeline leakage, bursts, and metering inaccuracies before reaching the end consumer—reaches an estimated 346 million cubic meters per day, which equates to over 126 billion cubic meters of high-quality freshwater wasted annually [3]. Beyond the direct loss of a precious resource, these systemic leakages accelerate localized geomorphic degradation, destabilize municipal soil beds, and generate enormous energy inefficiencies by inflating the pumping and treatment overheads required to compensate for lost volumes [4].

In the Mediterranean basin, which is warming at a rate significantly higher than the global average, water resources have assumed a fully strategic and geopolitical importance [2]. The region exhibits an average water stress index of 225.2%, compared to a global average of just 18.3% [2]. Within this vulnerable basin, Italy represents a highly critical case: it ranks as the first country in the Mediterranean area for value added generated by the extended water sector ($\approx$ \$31 billion) and exhibits an absolute lack of decoupling between Gross Domestic Product (GDP) and water consumption [2]. Indeed, water is the fundamental enabling resource for Italy's economy, supporting the generation of over **€384 billion in annual national value added** (representing approximately 20% of the Italian GDP) and over \$1.9 trillion across the broader Mediterranean region [2].

Despite the immense socio-economic value of water, Italy suffers from a profound, long-standing infrastructural debt. The national distribution network spans a massive **324,480 km of pipelines**, yet is characterized by an average physical water loss of **37.9%** in distribution, with regional losses regularly exceeding 40% in the central and southern districts [2]. On a national scale, this infrastructure disposes of an average linear loss of **24 m^3 of treated water per kilometer of network per day** [2]. This severe physical dissipation is directly tied to the obsolescence of the underground assets: among the portion of the network with recorded installation records (representing 44% of the total), over **30% of the active pipelines exceed 30 years of age**, thereby exceeding half of their standard engineering life expectancy and explaining the high rate of spontaneous pipe fractures [2].

This thesis focuses on Toscana, a region characterized by a highly consolidated and industrially mature water sector, but one that is increasingly exposed to extreme meteorological and geomorphic risks. During the 2025 hydrological year, Toscana experienced severe hydro-climatic stress: snowfall volumes plummeted by **45% below the 2010–2020 baseline**, and potential aquifer infiltration and groundwater recharge rates fell by **19% compared to the historical**

1991–2020 average [2]. Topographical hazards such as clayey soil compaction, landslides, and thermal soil expansions place severe mechanical stresses on underground conduits, leading to accelerated pipeline deformation and joint failures. To prevent severe structural failures and reduce Non-Revenue Water, the province of Firenze represents an ideal territorial laboratory to implement a proactive, digital diagnostic baseline.

This industrial need intersects with the technological paradigm of Industry 4.0, which has transformed modern utility management through cyber-physical systems (CPS), smart metering, and Low-Power Wide-Area Networks (such as LoRaWAN and NB-IoT) [5, 6]. These smart networks continuously stream massive quantities of high-frequency, multi-source, heterogeneous time-series data, recording precise flow and pressure dynamics across various nodes [7, 8]. Yet, a profound technological paradox persists in Italy's digital transition: while approximately 85% of the active networks have been georeferenced, the actual extension of distribution networks that are fully districtualized and equipped with active telecontrol systems **stands at a modest national average of only 38%** [2].

As a result, a substantial data utilization gap exists: municipal data streams are historically confined to retrospective logging, routine administrative billing, or basic reactive alarms [5]. In the era of Industry 4.0, data must be recognized as an abundant, non-rival, and inexhaustible asset—the "new oil" of the circular water economy—whose utility scales non-linearly when processed, synthesized, and integrated into advanced predictive models [9].

To bridge this data utilization gap, this project proposes a novel multi-source heterogeneous data fusion framework. By integrating localized, underutilized open-source heterogeneous environmental/socioeconomic data [10]—we can uncover latent structural correlations that are invisible in isolated systems [11, 12]. By shifting WDS management from a descriptive, reactive stance to a proactive, prescriptive paradigm, this framework enables the suggestion on pipeline deterioration, the localization of subterranean anomalies, and the optimized planning of preventive repairs [6], ensuring a resilient, efficient, and sustainable water transportation infrastructure for the province of Firenze and beyond.

1.2 Problem statement

Despite the recognized urgency of mitigating water loss and structural degradation in urban water distribution networks (WDNs), water utilities confront several systemic barriers that impede proactive infrastructure management. These challenges can be categorized into three interconnected problem domains: the structural data deficit, the operational and technological limitations of traditional diagnostics, and the institutional fragmentation of digital water data.

1.2.1 The Structural and Physical Data Deficit

A fundamental obstacle to modeling pipeline vulnerability is the chronic shortage, inaccuracy, or complete inaccessibility of in-situ physical asset data [13]. In many municipal networks—particularly legacy systems spanning rural and peri-urban corridors—exact installation records, pipe materials, joint configurations, and historical maintenance logs are either missing, unrecorded, or protected under strict security guidelines that limit their public dissemination [13, 10]. This structural opacity prevents the application of traditional, purely physical deterioration models that rely on pipe-level structural parameters to predict time-to-failure [14]. Without a comprehensive physical inventory, water utilities are left with a critical diagnostic blind spot, unable to assess which underground segments are undergoing active deformation or joint separation until a catastrophic rupture occurs.

1.2.2 Scalability Barriers in Traditional Diagnostics

To manage pipeline health, utilities have historically relied on localized, reactive, and contact-based physical inspection techniques, such as acoustic sensing, ground-penetrating radar (GPR), and manual surveys [4]. While highly accurate for local assessments, these methods are slow, labor-intensive, and fundamentally unscalable for extensive, regional-scale transmission networks [4]. They are ill-suited for the continuous, wide-area monitoring required to catch diffuse micro-leakages or early-stage structural anomalies before they escalate into high-volume losses [4, 10]. Consequently, underground leaks often persist undetected for months, eroding surrounding road beds, causing localized soil compaction, and driving up pump energy expenditures by forcing utilities to continuously pressurize compromised lines to meet user demand [4, 14].

1.2.3 Data Isolation, Fragmentation, and Semantic Heterogeneity

Concurrently on the data management side, a profound technological paradox exists: while municipal regions generate a massive volume of telemetric, geological, meteorological, and socioeconomic data daily, this resource remains severely underutilized [5, 12]. Municipalities and regional entities operate within highly fragmented data environments where valuable spatial and temporal streams are stored in isolated, discordant formats across distinct web services and databases [15].

This lack of coordination introduces severe semantic and syntactical heterogeneity, as different utility providers, public administrations, and environmental monitoring agencies employ highly divergent data structures, communication protocols, and variable definitions [11]. Standardizing these datasets into a unified model is historically slow and institutionally complex, resulting in a persistent reliance on ad-hoc databases and flat plain-text files that are entirely unsuited for large-scale analytical processing [11, 12].

1.2.4 The Gap in Integrated Decision Frameworks

Finally, the water sector lacks cohesive frameworks capable of translating scattered, multi-source data streams into prioritized, municipal-wide interventions [14]. In current administrative configurations, predictive modeling of pipe failure, hydraulic consequence simulations, and multi-criteria socioeconomic decision-making are treated as entirely independent, disconnected tasks [14]. Without an integrated, multi-source data fusion framework, regional water managers cannot identify latent correlations between external geomorphic risks (such as soil movement or extreme hydro-climatic events) and infrastructure vulnerability [13]. This analytical disconnect prevents utilities from transitioning from reactive, descriptive technical checks to a proactive, prescriptive maintenance paradigm, ultimately locking municipal entities into inefficient, localized, and retrospective operational structures [5, 14].

1.3 Research questions and objectives

To address the severe structural and data-driven bottlenecks characterizing municipal water distribution networks (WDNs), this research shifts the diagnostic focus away from highly restricted, in-situ physical asset parameters. Instead, it leverages the abundance of environmental, operational, and intellectual property data made available through modern public repositories. To systematically guide this investigation, a primary research question is formulated, supported by three interconnected secondary objectives.

1.3.1 Main Research Question

The core inquiry animating this thesis is formulated as follows:

How can multimodal open data and patent data be used to reduce inefficiencies in water transportation systems through pipelines?

This question seeks to establish whether unstructured and heterogeneous open-source registries—which are conventionally ignored by physical and hydraulic engineering models—can be successfully synthesized to reveal latent correlations between environmental stressors (causes) and infrastructural failure points (effects).

1.3.2 Secondary Research Objectives

To resolve the main research question, the project is divided into three consecutive, secondary operational objectives:

1. **Identification and Cataloging of Heterogeneous Repositories**

   To identify and systematically catalog diverse repositories of multimodal open data and global patent databases relevant to the Italian water transport system.

   The initial stage of this research addresses the critical variety of big data in the water sector [11]. It involves mapping and cataloging diverse public environmental registries (e.g., regional meteorological networks, soil moisture profiles, and geomorphic landslide hazards) alongside global patent databases (e.g., European Patent Office, WIPO) containing patented technological solutions for pipeline repair, leak prevention, and joint stabilization. High-variety datasets are crucial to compensate for the complete absence of in-situ structural measurements [13, 16].

2. **Methodology for Heterogeneous Integration and Interoperability**

   To establish a comprehensive methodology for the integration and harmonization of heterogeneous datasets, ensuring interoperability between various knowledge fields and from structured to unstructured data.

   Integrating multi-source, multimodal data introduces severe semantic and structural heterogeneity [11, 17]. Diverse data schemas, different spatial-temporal resolutions, and divergent variable definitions across regional administrative silos conventionally prevent unified analytical processing [11]. This objective aims to build an interoperable harmonization workflow—spanning spatial-temporal alignment and semantic mapping—capable of uniting unstructured text and spatial geodata into a cohesive database structure, bridging the historical telemetry deficit [5, 6].

3. **Design of a Cause-and-Effect Decision Framework**

   To design a data-driven framework that utilizes integrated insights built on causes and effects to support the prioritization of infrastructure interventions and the optimization of resource allocation. [13]

The final objective translates integrated data streams into actionable operational intelligence for water utility managers. By pairing external environmental vulnerabilities (causes) with localized patented mitigation technologies (solutions) [13], this objective designs a spatial decision framework. By moving beyond isolated hydraulic or statistical forecasts, the framework integrates multi-criteria decision support to help municipalities prioritize network rehabilitation pipelines, optimize maintenance budgets, and proactively schedule repairs under severe data limitations [14].

1.4 Structure of the thesis

To systematically address the research questions and operational objectives defined in this work, this thesis is organized into six logically sequenced chapters, reflecting a rigorous transition from initial problem formulation and multidisciplinary literature review to systematic data preprocessing, methodological modeling, technical implementation, and empirical validation. The structural roadmap of the document is aligned with the actual development of this project and is outlined as follows:

- **Chapter 1: Introduction**
  This introductory chapter establishes the foundational baseline and academic justification of the study. It opens with Section 1.1 (*Context and Motivation*), which positions the research within the global water crisis and Italy's contemporary hydro-climatic and economic realities using empirical data from the national service monitoring reports [2]. Section 1.2 (*Problem Statement*) then formulates the core systemic inefficiencies, categorized across four specific dimensions: the structural/physical data deficit in bulk water networks [13], localized scalability barriers in traditional acoustic and contact-based diagnostics [4], the administrative isolation and semantic heterogeneity of municipal databases [5], and the chronic absence of integrated geomorphic-hydraulic decision frameworks [14]. Section 1.3 (*Research Questions and Objectives*) defines the core boundaries of the inquiry, establishing a primary research question supported by three secondary operational objectives. Finally, Section 1.4 (*Structure of the Thesis*) provides this architectural roadmap.

- **Chapter 2: Literature Review & Related Work**
  This chapter conducts a comprehensive, multidisciplinary survey of the literature/state of the art in smart water infrastructure diagnostics, data integration and related fields which serve the industry 4.0. Section 2.1 (*Literature Review*) evaluates a historical pass into the paradigms present in digital water systems, cyber-physical systems (CPS), and other relevant systems for distribution modeling or risk prediction. Section 2.2 (*State of the Art*) explores contemporary technological developments, with Subsection 2.2.1 (*Application Context*) focusing on the field where these solutions where applied, how they could improve our case although they where not initially implemented in the water sector. Finally, Section 2.3 (*Research Gaps*) synthesizes the limitations of existing studies, highlighting the critical lack of scalable, non-invasive, and low-cost risk-modeling frameworks that can operate in the complete absence of in-situ telemetry data.

- **Chapter 3: Data Sources, Extraction, & Transformation**
  This chapter documents the systematic, catalog and data-centric ETL (Extract, Transform, Load) pipeline of the thesis, detailing how diverse open environmental registries are found, mapped, evaluated, and harmonized alongside industrial patent databases. To resolve the semantic and structural heterogeneity characterizing these distinct datasets, the

process is structured into eight consecutive operational phases, maintaining a dual-track exposition that contrasts spatial/environmental open data and socio economic data with the *Patent Case*:

  - *Data Source Discovery (Section 3.1 & Subsection 3.1.1 Patent Case):* Explores public environmental repositories (e.g., Copernicus, ISPRA, regional open data,meteoclimatic networks) and global patent databases (WIPO, EPO) to locate spatial and technological indicators.
  - *Data Source Identification (Section 3.2):* Specifies the precise sources, spatial resolutions, and temporal intervals of the target datasets in Toscana.
  - *Source Classification (Section 3.3 & Subsection 3.3.1 Patent Case):* Categorizes sources by their type, accessibility, diversity and governing entity.
  - *Data Source Evaluation (Section 3.4 & Subsection 3.4.1 Patent Case):* Conducts a quality-gate analysis to assess missingness, spatial granularity, and technological relevance.
  - *Extraction Methods (Section 3.5 & Subsection 3.5.1 Patent Case):* Details the programmatic scripts, web-crawling engines, and algorithms used to extract data from the repositories.
  - *Data Extraction (Section 3.6):* Executes the actual retrieval and compilation of raw tabular, spatial, and textual data.
  - *Data Characterization (Section 3.7 & Subsection 3.7.1 Patent Case):* Analyzes the mathematical and statistical properties of the extracted data, charting spatial distribution and patent term frequency trends.
  - *Data Transformation (Section 3.8 & Subsection 3.8.1 Patent Case):* Implements data cleaning, outlier imputation, and geospatial reprojection, transforming disparate inputs into a unified spatial format.

- **Chapter 4: Methodology**
  This chapter represents the core theoretical and modeling contribution of the thesis, delineating the mathematical formulation of the multi-source heterogeneous data fusion framework. Section 4.1 (*Overview*) introduces the conceptual paradigm. Section 4.2 (*Conceptual Pipeline*) details the construction of the dual-layer model: (1) the *Vulnerability Layer*, which leverages the Uber H3 hierarchical hexagonal spatial index to tessellate the province of Firenze and aggregate normalized open environmental covariates (e.g., slope, soil displacement, precipitation, and proximity to geomorphic hazards); and (2) the *Mitigation Layer*, which maps patented technological interventions (e.g., structural reinforcements, leak detectors, joint-stabilizers) to specific environmental risk factors. Section 4.3 (*Data Flow*) traces the computational and semantic transitions of the data across the model's layers. Finally, Section 4.4 (*Design Considerations*) addresses model assumptions, scalability constraints, and the trade-offs of excluding physical pipeline attributes.

- **Chapter 5: Implementation and Experimental Setup**
  This chapter details the technical and experimental environment built to validate the proposed geospatial data fusion model. Section 5.1 (*Implementation Overview*) describes the stack and system architecture. Section 5.2 (*Implementation Details*) documents the

concrete programmatic libraries (e.g., GeoPandas, h3-py, fiona) and database configurations used. Section 5.3 (*Experimental Setup*) details the pilot domain in the province of Firenze, Toscana. Section 5.4 (*Evaluation Protocol*) defines the spatial validation strategy, while Section 5.5 (*Experimental Scenarios*) configures distinct hydro-climatic stress levels (e.g., extreme drought-induced soil compaction versus heavy precipitation-induced land slippage). Finally, Section 5.6 (*Evaluation Metrics*) specifies the quantitative indices used.

- **Chapter 6: Results and Analysis**
  The final chapter presents and critically discusses the empirical findings of the study. Section 6.1 (*Overview*) summarizes the experimental outcomes. Section 6.2 (*Empirical Findings*) analyzes the spatial distribution of WDN vulnerability zones in Firenze, evaluates the precision of the H3 risk-prioritization algorithm, and showcases the automated pairing of localized geomorphic hazards with specific, patented engineering solutions. Section 6.2.1 (*Data Characteristics*) discusses the performance of the predictive model under varying granularities and evaluates the implications of the digitalization paradox on municipal asset management.

2 Literature review & related work

2.1 Literature review

The development of modern smart water distribution networks (WDNs) and the management of water loss representing physical distribution inefficiencies has emerged as a multi-disciplinary challenge. To solve these problems in a non-invasive, scalable, and computationally robust manner, contemporary literature has experienced a rapid convergence across several historically independent research domains. These include: (1) semantic Web and database technologies designed to resolve syntactic and conceptual data heterogeneity; (2) Industry 4.0 data management architectures supporting real-time cyber-physical integration; (3) remote sensing pipelines utilizing UAV, aerial, and satellite platforms for pipeline localization and leak identification; (4) machine learning algorithms applied to smart water metering and structural health monitoring (SHM); and (5) multi-criteria decision-support frameworks and closed-loop optimization systems designed to prioritize capital-intensive infrastructure renewals. To map the historical trajectory of these intersecting technologies, this section conducts a thorough, chronological review of the literature over the past two decades (2005–2026), detailing how early static data models have evolved into the contemporary closed-loop, data-driven, and physically consistent paradigms that ground this thesis.

2.1.1 The Foundations, Data Fusion and Predictive Maintenance (2005–2017)

The theoretical baseline of modern multi-source data integration began with the formalization of early semantic data schemas. [18] proposed a general multi-source data fusion framework based on the Resource Description Framework (RDF) and RDB-to-RDF Mapping Language (R2RML). This pioneering work demonstrated how relational databases (RDBs) containing structured physical coordinates could be programmatically mapped into unified RDF graphs, resolving semantic conflicts and schema heterogeneity through dynamic ontology mapping. By establishing rules to merge College, Paper, and Organization data models, this work proved crucial for subsequent efforts to break down institutional data silos, showing how semantic links can integrate unstructured and semi-structured Web data into an extensible knowledge base.

As industrial systems digitized in the subsequent decade, researchers recognized that traditional, centralized databases were ill-suited for the dynamic complexities of proactive asset protection. [19] proposed a systematic framework for the acquisition, structuralization, and characterization of multi-source heterogeneous industrial big data in the context of predictive maintenance. Crucially, Yan et al. introduced the concept of modeling "multiple invisible factors"—including environmental temperature, relative humidity, personnel capabilities, workspace noise, and weather forecast data—and correlating them with physical degradation metrics (such as tool cutter wear or bearing vibrations). By applying the Hilbert-Huang Transform (HHT) and Empirical Mode Decomposition (EMD) to decompose raw sensor streams, combined with Artificial Neural Networks (ANNs) for error backpropagation, this methodology proved that integrating context-aware, non-physical variables into machine learning algorithms could achieve superior accuracy in remaining useful life (RUL) estimation compared to unimodal, purely physical models. This established the foundational paradigm of utilizing environmental and external indicators to predict the structural health of physical assets, a methodology adapted in this thesis.

2.1.2 Cyber-Physical Systems and Intelligent Water Telemetry (2018–2020)

By the late 2010s, the deployment of cyber-physical systems (CPS) and the Internet of Things (IoT) triggered an exponential surge in telemetric data, requiring new distributed architectural designs. [5] conducted a horizontal survey of data management as it applies to networked industrial environments. Raptis et al. extracted key data properties (volume, variety, traffic, and criticality) across diverse industrial use cases, illustrating the transition from routine process log checking to localized, distributed computing. Their taxonomy divided the Industry 4.0 ecosystem into data enabling technologies (IIoT, WSANs, and networked control systems) and data centric services (anomaly detection, fault diagnosis, job scheduling, and ontologies based on IEC 61850 and IEC 61499), highlighting that edge and fog computing layers are essential to handle communication delays and reduce central processing overheads.

Concurrently, the concept of the digital twin (DT) was formalized to bridge the physical and virtual worlds. [20] proposed a 5-dimension cross-industry digital twin application model, which defined a DT as the integrated convergence of a physical entity, a virtual model, a centralized or distributed data center, a set of services, and the connections between them. Tao et al. emphasized that the "twin data" layer must deal with multi-temporal, multi-dimensional, and highly heterogeneous observations, utilizing semantic translation and XML encoding to synchronize virtual simulations with physical context. This digital twin theory was further expanded by [21], who mapped the progressive stages of industrial digitalization from digital enablement (converting paper records to flat text) to cyber-physical integration, categorizing the specific hardware, software, and simulation tools (such as MATLAB, ANSYS, and Azure IoT Hub) that support predictive maintenance and prognostics.

These macro-level industrial architectures were quickly adapted to municipal water supply and distribution networks. [12] conducted a systematic review of machine learning (ML) and data analytic (DA) techniques applied to residential digital water metering. Rahim et al. categorized the literature into five core themes: water-use feedback (web portals vs. paper billing), water event categorisation, water demand forecasting, behavior analysis, and socioeconomic analysis. Their critical review revealed that while the disaggregation of water end-use events (e.g., separating shower, toilet, and washing machine events) using k-means or CURE clustering had reached up to 93% accuracy, water utilities faced a severe "data warehouse gap," with massive high-frequency telemetric streams from Digital Water Meters (DWMs) restricted to simple billing rather than active leak detection or personalized behavior profiling.

To operationalize these telemetry grids, [6] introduced the "SMART-WATER" solution, a unified five-layer platform designed for remote telemetry and remote-control of municipal water consumption. Antzoulatos et al. deployed smart automated meters and remote-controlled valves utilizing LPWAN protocols (specifically wM-Bus EN 13757-4, LoRaWAN, and NB-IoT), establishing a GSM-connected multi-protocol gateway (MPG) that successfully bypassed proprietary vendor lock-in. Their platform applied Auto-Regressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM) models to predict hourly urban water consumption in Thessaloniki, Greece, demonstrating that near real-time telemetry could identify background leakage anomalies before a catastrophic rupture occurred. This IoT-driven SCADA paradigm was mirrored by [15], who implemented a low-cost Supervisory Control and Data Acquisition (SCADA) method based on Raspberry Pi controllers for underwater operation safety and municipal water resource quality control. Nie et al. formulated water distribution networks as connected mathematical graphs, applying continuity rules of mass and energy conservation alongside spectral analysis of big data to calculate the highest frequency of water loss cycles, thus allowing the proactive control of pump settings to minimize background leakage.

The application of big data pipelines to assess regional environmental risks also experienced significant breakthroughs. [15] developed a national-scale big data analytics pipeline to assess the potential impacts of flooding on critical infrastructure. Donratanapat et al. designed the Flooding Impact Assessment System (FAIS), demonstrating how automated APIs could retrieve tabular, spatial, and image data from ground-based monitoring gauges, live traffic, and river webcams to dynamically map community and infrastructure vulnerabilities, bridging the gap between raw spatial data and actionable decision-making.

To process the structural and semantic heterogeneity of these multi-source datasets, researchers turned to graph neural networks and multimodal deep learning. [22] developed a Multi-Source Information Fusion based Heterogeneous Network Embedding (MIFHNE) approach, mathematically proving that jointly modeling structural proximity, attribute info, and label info within a Nonnegative Matrix Factorization (NMF) framework could capture semantic relations via meta-graph-based random walks, mapping complex, non-Euclidean spaces to uniform vector representations. This was horizontally supported by early Spatio-Temporal Graph Convolutional Networks (STGCNs) that aggregated irregular monitoring stations into fully connected graph-structured datasets to capture spatially correlated representations of regional environments [23]. Simultaneously, [24] surveyed deep learning models for multimodal representation, proving that deep neural networks (e.g., Inception, SEnet, and ECNN) could successfully learn joint distributions across highly diverse, unstructured modalities (such as images, text, and numerical vectors) by utilizing attention mechanisms and autoencoder representations to capture cross-modal correlations. [16] validated this paradigm by demonstrating that correlating financial and environmental time-series with micro-blogging Web activity could significantly reduce forecasting errors, showcasing the value of context-aware data fusion. On the physical asset management side, early national-scale databases began to compile historical collections of water pipeline field performance [25], establishing standard data models to estimate pipe likelihood of failure and analyze economic life-cycle costs under varying bedding conditions, soil properties, and external loads (such as live traffic, railways, and ground movement).

2.1.3 Semantic Web Integration, Structural Monitoring, and Patent Intelligence (2021–2023)

As the Industry 4.0 paradigm matured, data was recognized as a non-rival, inexhaustible resource—the "new oil" of the modern circular economy [9]. Klingenberg et al. conducted a

systematic review of 119 articles, categorizing technologies into Data Generation, Data Transmission, Data Conditioning/Storage, and Data Application (which creates the actual value), emphasizing that vertical and horizontal integration are essential to establish reliable data flows across value chains. However, the sheer variety of data generated by disparate municipal, environmental, and geological monitoring agencies introduced severe syntactic and conceptual heterogeneity. [11] addressed these challenges, demonstrating how the DIKW (Data, Information, Knowledge, Wisdom) pyramid could be realized by utilizing Knowledge Graphs in combination with ontologies (such as SensorML, Optique, and SAREF4INMA) to semantically integrate unstructured and heterogeneous databases. Kamm et al. proposed a unified concept to perform data mining on heterogeneous datasets through a two-stage process: first, using feature extraction to reduce the dimensionality of separate data types (images, time-series, text), and second, feeding these features into ensemble learning models (such as majority voting or weighting) to enable "end-to-end" learning.

This semantic digitalization led to localized digital water software designs, such as the Water Wise software system (W2S) for smart cities and water management entities [3]. Figueiredo et al. presented a preliminary study of W2S, which integrated SCADA, GIS, and EPANET with machine learning and deep learning (Support Vector Machines and LSTMs) to optimize the water-energy nexus. W2S utilized Kafka streaming and Mulesoft Enterprise Service Bus to process high-frequency data from District Meter Areas (DMAs), allowing water utilities to classify consumers, identify consumption anomalies, and prevent service interruptions in real time. On the hardware monitoring front, [26] conducted a comprehensive review of machine learning pipelines in structural health monitoring (SHM). Malekloo et al. outlined the 5-step hierarchical damage identification scheme (Level 1: detection, Level 2: localization, Level 3: classification, Level 4: severity, and Level 5: prognosis) and recommended a unified pipeline consisting of contact-based or vision-based data acquisition, data normalization (GMM, KCPA), data cleaning/imputation (LSTM, CNN), data compression (PCA, autoencoders), data fusion, and pattern recognition (Bayesian and deep NNs). Crucially, the authors reviewed population-based SHM, which transfers damage-sensitive knowledge between similar and dissimilar structures, addressing the chronic scarcity of damaged training labels under environmental and operational variations (EOFs).

This era also marked the birth of advanced patent intelligence pipelines, which sought to mine global patent databases to track and extract specific engineering solutions. [27] introduced a novel methodology combining Named Entity Recognition (NER) and dynamic, time-varying network analysis to measure technological convergence across more than 300,000 patents. By applying terminology-based, Bert-boosted, and rule-based NER to extract technical keywords from unstructured corpora, Giordano et al. constructed yearly co-occurrence networks and applied weighted modularity algorithms to mathematically track the birth, stability, merging, and splitting of distinct technologies over time. To extract data from the graphical portions of patent documents, [28] developed a deep learning pipeline to segment, detect, and map technological keywords from patent drawings and block diagrams. Using Python OpenCV contours and Optical Character Recognition (OCR), Jee et al. extracted keywords from in-diagram rectangles and constructed a keyword-to-keyword (K2K) matrix. By assuming that the topological distance between keywords in block diagrams serves as a proxy for physical or functional relationships, this work established a technology structure map that successfully grouped components (Encoder, Decoder, CPU, DSP, Gateway) without the noise inherent in full-text patent descriptions. This graphical mining was horizontally supported by [29] and [30], who focused on multi-modal representation, using Grabcut region segmentation, Fourier Descriptors, and Support Vector Machines (SVMs) to extract spatial, geometric, and topological

shapes from patent drawings to identify joint-stabilization and pipe-repair innovations.

In parallel, IoT-driven water quality monitoring became highly standardized. [7] conducted a systematic review of low-cost IoT smart water management systems, detailing how ESP8266 or Arduino controllers could continuously transmit pH, turbidity, salinity, dissolved oxygen, temperature, and conductivity parameters to cloud databases, allowing real-time water resource monitoring with minimal cost. Additionally, the geomorphic modeling of soil and water hazards accelerated. [31] proposed hybrid deep learning architectures (CNNs, Vision Transformers, and LSTMs) for spatial-sequential risk prediction of floods and landslides, while [32] reviewed multi-temporal remote sensing advances to predict soil texture profiles, which directly influence the mechanical settlement of underground pipelines. Multitemporal remote sensing became widely utilized to map buried pipelines and soil anomalies [33]. John S. Lioumbas et al. utilized multitemporal Sentinel-2 satellite imagery (2015–2025) and DJI Mavic 3 high-resolution drone imagery to detect a 52-kilometer subterranean water aqueduct in Thessaloniki, Greece. By analyzing "soil marks"—perceptible alterations in soil color, moisture, and thermal properties radiating from the pipeline trench and backfill material—the authors successfully delineated the pipeline route as a 20-to-25-meter-wide surface proxy, disentangling pipeline-induced anomalies from seasonal vegetation cycles.

To evaluate the operational consequences of pipeline failure, [34] introduced probabilistic frameworks to assess network influence and hazard zones under uncertain operating conditions, allowing water utilities to simulate cascading pressure drops. At the treatment and reuse scale, [35] explored risk management and digitalisation to overcome regulatory and social barriers for wastewater irrigation in European and Italian practice, integrating conventional sensors with innovative cyber-physical networks to ensure compliance with ISO 16075:2020 and JRC guidelines. To process the non-linear, multi-source correlations inherent in these systems, water utilities borrowed advanced multi-source graph neural network algorithms and text-mining models originally designed for tracking highly dynamic and complex financial markets [36, 37]. These financial graph architectures aggregated textual news and numerical transactions into heterogeneous subgraphs with node-attention and edge-weight matrices, demonstrating how text-image relations could be fused to predict complex, non-linear system volatilities.

2.1.4 Scalable Remote Sensing, Physics-Informed ML, and Closed-Loop Control (2024–2026)

In recent years, the literature has experienced a major shift towards non-invasive spatial diagnostics and physics-informed machine learning. [4] achieved a breakthrough in leak detection by integrating L-band Synthetic Aperture Radar (SAR) satellite imagery from ALOS-2 with machine learning. Utilizing dual-polarized SAR images (HH and VV modes, 6-meter spatial resolution) and Gray-Level Co-occurrence Matrix (GLCM) feature extraction, Saleh et al. associated historical leak reports from the Hong Kong Water Supplies Department (WSD) within a 30-day temporal window. Their stacking ensemble algorithm achieved an overall pixel classification accuracy of 80%, demonstrating the scalability of detecting underground moisture anomalies in high-density urban environments without requiring in-situ physical inspections. This was complemented by the landmark review of [38], who surveyed the transformation of smart manufacturing through generative AI, large language models (LLMs), and sensing-AI codesign. Gao et al. emphasized that to build robust and trustworthy systems, researchers must replace uninterpretable, purely data-driven black-box neural networks with physics-informed layers (e.g., continuous wavelet layers) to enforce physical consistency and domain constraints. For localized structural convergence monitoring, [39] proposed distributed optical fiber strain sensors (DOFS) based on stimulated Brillouin scattering (SBS) and Rayleigh backscattering. By

externally mounting optical fiber cables on high-density polyethylene (HDPE) water pipelines.

To handle the real-time querying of these massive multi-source datasets, [17] proposed "FusionQuery," an on-demand VLDB framework designed to resolve between-source conflicts on-the-fly using only query-related data. Zhu et al. formulated the heterogeneous query problem as a knowledge graph matching problem. To bypass the exponential search space of traditional graph matching, FusionQuery introduced a "knowledge line graph transformation" to decouple semantic information from graph structure, reducing the matching complexity to $O(|R_q||R_d|)$ and utilizing an Expectation-Maximization (EM)-style algorithm to iteratively estimate data veracity and source trustworthiness. Simultaneously, real-time edge monitoring platforms such as "HydroSense 2.0" [40] integrated an embedded control layer (ESP32) housing pH, turbidity, TDS, and temperature sensors inside a low-cost PVC collar joint with a PythonAnywhere cloud analytics layer, utilizing machine learning algorithms to send instant SMS and web dashboard leakage alerts. This was supported by watershed automatic annotation methods [41] and the work of [42], who conducted a PRISMA-compliant systematic review of AI-driven bathing water quality forecasting models, evaluating the capabilities of Google Earth Engine and Sentinel-2 satellite images for regional water quality and environmental monitoring. At the secure governance scale, [1] systematically reviewed the integration of blockchain with ML/DL models from 2019 to 2025, demonstrating how smart contracts and decentralized frameworks could secure groundwater levels, water rights trading, and wastewater quality compliance.

To map underground pipeline networks accurately in rural environments, [10] developed an integrated, multi-component remote-sensing workflow that integrated UAV-borne near-infrared (NIR) surveys, multi-temporal Sentinel-2 satellite imagery, and historical Google Earth orthophotos. By processing each dataset within a unified least-squares framework, Lioumbas et al. delineated pipeline axes from surface anomalies (vegetation stress, soil discoloration, and proxies) and rigorously quantified positional uncertainty (E_{total}), achieving sub-meter accuracy (± 0.3 m) with UAV data, meter-scale precision ($\approx \pm 1$ m) with Google Earth, and several meters precision (± 13.0 m) with Sentinel-2. This methodology demonstrated that optimal UAV-based identification occurs during early-vegetation growth phases (NDVI $\approx$ 0.30–0.45) and post-harvest periods, successfully guiding excavation to locate a pipeline segment that diverged by over 300 meters from inaccurate legacy maps. In parallel, advanced ensembling techniques saw significant upgrades. [43] proposed a novel approach utilizing Directed Acyclic Graphs (DAGs) to dynamically structure the forecasting flow of multiple machine learning classifiers (SVM, RF, CNN, and XGBoost). By modeling inter-classifier dependencies through directed arcs (where model outputs serve as inputs to subsequent models) without circular reasoning, their DAG ensemble framework successfully combined horizontal and vertical integration, achieving superior prognosis and remaining useful life (RUL) estimation on complex sensor datasets compared to static stacking or voting. This was supported by general secure anomaly detection frameworks [8] and modern multi-source heterogeneous data fusion models [16] designed to handle highly correlated streams.

To bridge the gap between risk prediction and operational execution, the latest literature has focused on integrating multi-criteria decision frameworks and physics-informed closed loops. [14] presented a pioneering framework integrating machine learning (Random Forest, XGBoost, and Logistic Regression) with Multi-Criteria Decision Analysis (MCDA, specifically TOPSIS) to perform spatial risk ranking and maintenance prioritization in drinking water networks, accounting for environmental, geological, and socio-economic hazards. By applying TOPSIS with bootstrap resampling of expert weights (risk reduction, executability, reliability, cost, social concern), Sangroula et al. successfully prioritized interventions (pressure management, pipe repair/rehabilitation, pipe replacement), validating the robustness of the rankings against

the Weighted Sum Model (WSM) via Spearman's rank correlation ($\rho = 0.85$). Finally, [44] systematically synthesized the past decade of AI in water management, proposing a unified, deployable closed-loop "monitor-predict-optimize-control" architecture. Yiting et al. emphasized the transition from purely data-driven black-box neural networks to Physics-Informed Neural Networks (PINNs) and modular graph neural networks that embed activated sludge conservation laws and operational constraints directly into learning algorithms, coupled with Multi-Objective Reinforcement Learning (MORL) to minimize energy and carbon footprints during proactive maintenance. Fusing these predictive, multi-criteria, and patent-mitigation streams allows the operationalization of regional water transportation plans, directly aligning with the latest Italian water service monitoring indicators documented in the national Blue Book reports [2].

2.2 State of the art

The modern management of municipal Water Distribution Networks (WDNs) has undergone a major paradigm shift, transitioning from localized, manually intensive, and reactive leak-detection methods to continuous, non-invasive, and cyber-physically integrated diagnostics. These contemporary advancements, represent the state-of-the-art solutions in smart water infrastructure. By leveraging satellite observation, UAV-borne multispectral sensors, advanced telemetry, and physics-informed modeling, these inventions in the literature are some of the most current and serve as the state of the art for future improvements or innovations in the water transportation fields.

2.2.1 Non-Invasive Subterranean Pipeline Mapping and Leak Detection

A primary challenge in regional water transport systems is localizing buried pipeline corridors and identifying underground leak anomalies without extensive, blind excavations. Over the last years, remote sensing has emerged as a key solution to this problem, leveraging soil-moisture and vegetation-stress spectral proxies.

- **Multi-Temporal Satellite and UAV Fusion:** Delineating subsurface aqueducts in rural environments historically suffered from high positional uncertainty. To resolve this, established workflows integrating multi-temporal Sentinel-2 multispectral imagery, historical Google Earth orthophotos, and low-altitude Unmanned Aerial Vehicle (UAV) near-infrared (NIR) surveys [10].
- **L-Band SAR and Ensemble Stacking:** In paved, dense urban environments where optical spectral signatures are obstructed, utilizing active satellite radar. [4] demonstrated that L-band Synthetic Aperture Radar (SAR) from the ALOS-2 satellite can penetrate urban pavement and topsoil layers to capture subsurface dielectric variations caused by pressurized water pipe leaks.

2.2.2 Bidirectional Telemetry, Edge Computing, and Digital Water Software

The physical pipeline layer is mirrored in the cyber domain through real-time communication architectures and cloud-based analytics engines that bridge the digitalization gap.

- **Bidirectional Multi-Protocol Telecontrol:** The "SMART-WATER" platform represents a major milestone in fixed wireless telemetry, deploying custom bidirectional and multi-protocol gateways capable of simultaneously decrypting and managing data streams

from wM-Bus, LoRaWAN, NB-IoT, and GSM end-devices [6]. This platform enables water utilities to perform synchronous hourly telemetry of consumption patterns and dynamically actuate wireless water valves to isolate burst pipes in real time.

- **Energy-Autonomous Smart District Metering:** Building on the concept of District Metered Areas (DMAs), the "Water Wise System" (W2S) software architecture integrates SCADA, GIS, and EPANET hydraulic modeling with machine learning classifiers (SVM) for real-time leak disaggregation and demand forecasting [3].

2.2.3 Integrated Decision Support and Closed-Loop Control Frameworks

To translate these diverse sensory inputs into targeted, plant-wide interventions, the latest literature has advanced toward integrated decision-support and physics-informed closed-loop control.

- **Multi-Criteria Risk Prioritization:** To resolve the historic disconnect between predictive failure modeling and administrative planning, current contributions integrate machine learning break-probability classifiers (XGBoost, Random Forest) with Multi-Criteria Decision Analysis (MCDA) frameworks, specifically the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [14]. Tested on operational networks, this framework ranks physical and hydraulic consequences alongside human health hazards using Quantitative Microbial Risk Assessment (QMRA) to systematically prioritize capital-intensive pipeline rehabilitations under tight utility budgets.

- **Physics-Informed Closed-Loop Systems:** Current smart water engineering lies in the transition toward closed-loop "prediction-optimization-control" architectures [44]. Differentiating themselves from early, black-box data-driven models, these systems employ Physics-Informed Neural Networks (PINNs) and modular graph neural networks (GNNs) that embed mass conservation laws, hydraulic head constraints, and activated sludge kinetics directly into the neural network architecture.

2.2.4 Application context

The formulation of a multi-source heterogeneous data fusion framework for water distribution systems (WDS) requires a thorough understanding of the domain-specific environments from which its technological and methodological building blocks are drawn. Rather than operating in an academic vacuum, the literature surveyed in this thesis spans five primary fields of engineering and applied science. This subsection analyzes and defines the specific application contexts, operational constraints, and technical parameters of the inventions documented in the bibliography, detailing how their respective principles converge to support the proposed water transportation predictive model.

Urban Water Engineering and Hydro-Informatics (Water 4.0)

Urban water management, water safety planning, and intelligent water telemetry. In this field, inventions focus on transitioning municipal water distribution networks (WDNs) from static, reactive utilities into active, cyber-physical systems.

- **Utility Performance and Sectoral Auditing:** The national baseline is established by Fondazione Utilitatis and Utilitalia [2], who compile and analyze Italy's integrated water service indices.

- **Digital Water Management Systems:** Figueiredo et al. [3] introduce the "Water Wise System" (W2S), an R&D software platform that targets real-time monitoring and decision support.
- **Bidirectional LPWAN Telemetry:** Antzoulatos et al. [6] design and deploy the "SMART-WATER" solution in Thessaloniki, Greece.
- **Low-Cost IoT Architectures:** Singh and Ahmed [7] conduct a systematic review of low-cost IoT smart water management systems.
- **Predictive Edge Monitoring platforms:** Mattihalli et al. [40] design and deploy "HydroSense 2.0", a low-cost, portable, and cloud-integrated early warning system.
- **Digital Water Metering and Demand Forecasters:** Rahim et al. [12] conduct a systematic review of machine learning and data analytics applied to residential digital water metering data.
- **Underwater SCADA Telemetry:** Nie et al. [15] implement a real-time SCADA framework specifically for underwater operations and municipal water networks.
- **Centralized Forensic Registries:** The Enterprise GIS Team at Virginia Tech [25] outlines the "PIPEiD" platform, a centralized water pipeline infrastructure database designed to serve utilities, regulators, and modelers.
- **Probabilistic Hydraulic Modeling:** Bado et al. [34] formulate a probabilistic framework to calculate the spatial boundary of water network influence and pressure hazard zones.
- **Decentralized Water Governance:** Mithal and Thankachan [1] systematically survey the integration of blockchain ledger architectures (such as Smart Contracts) with machine learning from 2019 to 2025.
- **Reclaimed Irrigation Compliance:** Radini et al. [35] focus on risk management and digitalization within European agricultural reclamation.
- **Closed-Loop Biological Process Control:** Xie et al. [44] synthesize the state of the art in smart wastewater treatment plants (WWTPs), detailing the transition from static, steady-state hydraulic models to closed-loop "prediction-optimization-control" frameworks.
- **Multi-Criteria Risk Management Systems:** Sangroula et al. [14] demonstrate an integrated health risk assessment framework under the World Health Organization (WHO) Water Safety Plan (WSP) guidelines in Sweden.

Earth Observation, Geospatial Hydrology, and Remote Sensing

When physical asset registers are missing or protected under strict security guidelines, researchers apply remote-sensing inventions to map terrain and subterranean pipelines non-invasively.

- **Subterranean Soil Mark Identification:** John S. Lioumbas et al. [33] explore the detection of a 52-kilometer subterranean water concrete aqueduct in Thessaloniki, Greece.

- **Multi-Component Remote Sensing Mapping:** Lioumbas et al. [10] expand this workflow by developing an integrated, multi-source pipeline that fuses UAV-borne near-infrared (NIR) surveys, Sentinel-2 imagery, and historical Google Earth orthophotos within a unified least-squares framework.
- **L-Band SAR Leak Detection:** Eslam Ali Saleh et al. [4] achieve a breakthrough in non-invasive, wide-area diagnostics by utilizing L-band Synthetic Aperture Radar (SAR) imagery from ALOS-2.
- **Pedological Classification and Modeling:** Pedological soil databases are evaluated by remote-sensing practitioners to predict soil texture profiles [32], which directly influence pipeline settlement risks.
- **Natural Hazard Prediction:** Singh et al. [31] propose hybrid deep learning architectures (CNNs, Vision Transformers, and LSTMs) for spatial-sequential prediction of floods and landslides.
- **Environmental Water Quality Monitoring:** Khan et al. [42] survey AI-driven bathing water quality forecasting models, evaluating Google Earth Engine and Sentinel satellite images to track regional coastal and surface water parameters.

Industrial Cyber-Physical Systems and Mechanical Diagnostics

The structural modeling of pipeline deterioration borrows heavily from industrial engineering, condition monitoring, and machine-level diagnostics.

- **Invisible Factors Modeling:** Jihong Yan et al. [19] propose a pioneering Industry 4.0 framework that models "multiple invisible factors"—such as ambient temperature, humidity, noise, and vibration—alongside physical metrics.
- **Industrial Data Management Taxonomy:** Raptis et al. [5] horizontal survey data management in networked industrial environments, extracting data properties (volume, variety, traffic, and criticality) across the automation pyramid.
- **Functional Data Classification Frameworks:** Klingenberg et al. [9] classify 111 Industry 4.0 technologies into four distinct groups (generation and capture, transmission, conditioning/storage, and application), showing that data acts as a non-rival, value-increasing asset.
- **Sensing-AI Codesign and Wavelet Nets:** Robert X. Gao et al. [38] review the integration of generative AI and large language models (LLMs) in manufacturing.
- **Secure Anomaly Detection:** Baimukhanov et al. [8] and Sultan Baimukhanov et al. [8] propose distributed anomaly detection schemes.
- **DAG-Based Ensemble Learning:** Ghosh et al. [43] propose a novel predictive maintenance model utilizing Directed Acyclic Graphs (DAGs) to structure the forecasting flow of multiple machine learning classifiers (SVM, Random Forest, CNN, and XGBoost).
- **Digital Twin Software Architectures:** Fei Tao et al. [20] and Qi and Tao [21] formalize 5-dimensional digital twin classification models.

- **Acoustic and Optical Structural Monitoring:** Konstantinou et al. [39] review distributed optical fiber strain sensors (DOFS), detailing how Brillouin optical time-domain analysis (BOTDA) achieves sub-millimeter strain resolution to detect micro-deformations.

Patent Intelligence, Technological NLP, and Multimodal Semantic Systems

To translate localized geological hazards into engineering interventions, semantic data fusion, technological natural language processing (NLP), and patent mining algorithms.

- **Dynamic Patent Convergence Networks:** Vito Giordano et al. [27] introduce a novel methodology that applies terminology-based, BERT-boosted, and rule-based Named Entity Recognition (NER) to extract technical keywords from unstructured corpora.
- **Graphical Keyword Extraction from Drawings:** J. Jee et al. [28] develop a patent analysis method focusing on block diagrams.
- **Multimodal Image Retrieval:** Kader Pustu-Iren et al. [29] and Li et al. [30] focus on semantic patent image retrieval.
- **On-Demand Database Queries:** Junhao Zhu et al. [17] propose "FusionQuery," an on-demand Very Large Database (VLDB) query framework.
- **RDF Schema Standardization:** Zhang et al. [18] propose a general multi-source data fusion framework based on Resource Description Framework (RDF) schemas.
- **Semantic Unstructured Integration:** Simon Kamm et al. [11] address the data variety bottleneck in Industry 4.0, detailing how Knowledge Graphs and feature extraction can semantically integrate unstructured databases to enable "end-to-end" machine learning.
- **Big Data Fusion Concepts:** Lili Zhang et al. [16] and Zhang et al. [18] analyze the theoretical foundations, data dimensions, and alignment protocols of heterogeneous multi-source fusion models under varying cloud architectures.

Relational Graph Embeddings and Financial Market Engineering

Finally, fields like financial merket engineering have developed prediction models ingesting heterogeneous sources.

- **Financial Graph Neural Networks:** X. Li et al. [36] and Jun Wang et al. [37] develop GNN stock forecasting methods based on heterogeneous human produced sources, i.e social media.
- **Spatio-Temporal GCN Embeddings:** Bentian Li et al. [22] and B. Li et al. [23] propose multi-source information fusion algorithms based on heterogeneous network embeddings (MIFHNE).

2.3 Research gaps

While the literature has extensively investigated advanced methodologies for water distribution network (WDN) diagnostics and structural health monitoring, a critical evaluation of the state of the art reveals a fundamental operational bottleneck. Traditional deterioration and risk modeling paradigms remain heavily dependent on highly restricted, proprietary physical and structural asset parameters—such as precise pipe materials, wall thickness, joint configurations, and historical in-situ breakage logs. Access to these datasets is strictly controlled by water utilities and private operators, creating an entry barrier for external researchers, public environmental agencies, and regional policymakers.

Concurrently, while highly precise, non-invasive diagnostic solutions such as Synthetic Aperture Radar (SAR) or multi-component UAV near-infrared (NIR) mapping have emerged, their practical execution still proves to be highly specific, thus having difficulties scaling the studies.

This research addresses this diagnostic bottleneck by focusing on an unaddressed gap in the literature: the formulation of a highly flexible, open-data, non-invasive risk assessment framework that can operate in the *absence* of proprietary structural pipeline attributes.

Rather than attempting to reconstruct high-fidelity physical pipe profiles or reconcile fragmented utility registries, this thesis pivots the modeling dimension entirely to publicly accessible, environmental, geomorphic, and meteorologic open-source covariates [13]. By tessellating the territory of Toscana into a unified geospatial index (such as Uber's H3 hierarchical hexagonal grid) and aggregating open datasets (e.g., multi-temporal satellite soil moisture anomalies, geomorphic landslide inventories, and historical precipitation models).

Although this non-invasive, open-source approach lacks the granular mechanical information of in-situ acoustic logs or private utility records, it offers flexibility and accessibility. Because it relies exclusively on public, remote sensing, and open-source data streams, the proposed risk-modeling methodology can be independently deployed and executed by any agency, public policymaker, or third-party conservation group outside of the private utilities.

Furthermore, this thesis bridges the gap between risk profiling and technological mitigation by pairing geospatially indexed open risk zones with global patent databases. By automatically identifying and mapping patented mechanical and monitoring solutions to specific environmental stressors (such as matching landslide hazard zones with patented flexible-joint technologies), this work provides regional water authorities with a transparent, low-cost, and non-proprietary tool to prioritize infrastructure funding and formulate regional water preservation plans under severe data limitations.

3 Data Sources, Extraction, & Transformation

3.1 Data Source Discovery

The initial phase of this research focused on an exploratory, manual web-based search strategy to discover and map public data repositories concerning Italian water transport and distribution networks [45]. In Italy, water infrastructure is managed in a highly decentralized manner by a variety of regional authorities, inter-municipal consortia, and private or public utility operators [46, 47, 48]. Consequently, establishing a baseline data directory required manually searching open portals using key target terms, specifically: **"rete idrica" (water network)**, **"acquedotto" (aqueduct)**, and **"servizio idrico integrato" (integrated water service)** [45].

3.1.1 Exploration of Repositories and Portals

The manual discovery process began by cataloging macro-level water statistics and spatial base layers from European, national, and regional repositories:

- **European Databases (Eurostat & Copernicus):** At the European scale, Eurostat was explored for baseline national water resources [49, 50], pipeline transport indicators, and regional geodata grids (GISCO) [51, 52]. However, Eurostat revealed zero river basin-level records for Italy, as well no pipeline infrastructure information [49, 50] nor sewage sludge datasets for the country [51, 52]. Environmental covariates were identified via the **Copernicus Data Space Ecosystem**, including ERA5 hourly climate reanalysis grids (for precipitation and temperature) which detail a live source and not a historic [53, 54] and Sentinel-2 crop classification (CLMS) datasets [55, 56].
- **National Statistical & Regulatory Repositories (ISTAT & ARERA):** National data discovery relied heavily on ISTAT and ARERA. ISTAT provides long-term statistics on public water supply abstractions (PRELACQ 1) [57, 58], municipal distribution volumes and loss rates (CONSACQUA 1, 2, 3) [57, 58, 59], wastewater treatment capacities (IMPDEP 1, 2) [57, 59], and municipal-level landslide and flood hazard indicators [60]. ARERA, the national regulator, manages economic datasets on Integrated Water Service (SII) expenditures [61] and operator compliance with technical and contractual quality standards (RQSII) [61, 62, 63]. ARERA annually validates and transmits national water utility performance datasets to ISPRA, including linear and percentage water losses (Indicator M1), delivered water quality (Indicator M3), and sewerage adequacy (Indicator M4) [64, 65].

National Geodata Registry (GeoDati - RNDT): The *Repertorio Nazionale dei Dati Territoriali* (RNDT) served as the primary catalog for discovering GIS layers, accounting for **47.64%** of all manually identified data resources in the initial catalog. The RNDT portal facilitated the discovery of regional vector layers such as state-owned public water domains [66], CTRN 5000 hydrographic elements and flow nodes [66], Strahler stream order classifications (DBGT 10K) [67], and official Water Framework Directive (WFD) monitoring stations [68, 69].

3.1.2 Automated Metadata Harvesting and API Discovery

To overcome the manual constraints and potential selection bias of human portal exploration, the second phase of the data discovery process implemented an automated metadata harvesting

architecture. The goal of this phase was to systematically query, collect, and index all public open data metadata pertaining to water transport networks, hydraulic infrastructure, and environmental risk covariates across Italian public administration repositories [70, 71].

The automated pipeline targeted three primary administrative tiers:

1. The national open data catalog (`dati.gov.it`), acting as the primary national aggregator [45, 72].
2. Regional open data portals across all 20 Italian administrative regions and autonomous provinces [71].
3. National regulatory and statistical portals, including the *Autorità di Regolazione per Energia Reti e Ambiente* (ARERA) data services [46, 73].

API Search Architecture and Query Parameterization

Automated harvesting was executed by interfacing directly with standardized Application Programming Interfaces (APIs). Portals deploying the Comprehensive Knowledge Archive Network (CKAN) platform were queried via the `package_search` API endpoint [70], while statistical repositories were queried using Statistical Data and Metadata eXchange (SDMX) REST interfaces [73].

Table 1: Exclusion Criteria for Regional Open Data Portals in Automated Harvesting

Exclusion Category	**Technical Cause**	**Impact on Harvesting**
Non-Standard APIs	Portal lacks CKAN, SDMX, or OGC/CSW endpoints	Inability to execute structured queries
Access Restrictions	Mandates proprietary authentication or CAPTCHA	Blocked programmatic harvesting
Unstructured Catalogs	Metadata stored solely in static PDF/HTML pages	Lack of machine-readable schema
Rate Limiting / Timeouts	Aggressive HTTP throttling without retry headers	Incomplete catalog responses

A critical requirement of the automated harvesting methodology was protocol standardization. Certain regional open data portals were explicitly excluded from the automated pipeline (Table 1) because they failed to support common interoperability standards—such as CKAN REST endpoints or OGC Catalog Services for the Web (CSW)—or imposed proprietary authentication barriers. Excluding non-standard portals ensured that the collected metadata maintained structural consistency and schema comparability [70, 71].

To capture all relevant datasets across the entire lifecycle of water transport, supply, drainage, and risk, a comprehensive array of 21 Italian search terms (`WATER_QUERIES`) was programmatically submitted to each API endpoint:

$$\begin{aligned}\texttt{WATER_QUERIES} = \{&\text{'acqua', 'acquedotto', 'fognatura', 'depurazione', 'tubazioni',}\\ &\text{'condotte', 'condotta', 'conduttura', 'serbatoi', 'serbatoio',}\\ &\text{'impianti', 'idrico', 'idriche', 'idrografico', 'bacini',}\\ &\text{'portata', 'pressione', 'dissesto', 'PGRA', 'fluviali', 'suolo'}\}\end{aligned} \tag{1}$$

Macro-Regional Disparities and Territory Distribution

The automated mass fetching operation yielded a consolidated catalog of **10,922 metadata records** [71]. Analysis of this harvested metadata reveals an extraordinary geographic disparity in open data publication across Italian territories (Table 3 and Table 2). (altough noting some of the regional portals were excluded)

Table 2: Metadata Extracted per Macro-Area from Automated API Harvesting [71]

Macro-Area	**Harvested Count**	**Percentage (%)**
Centre	5,497	50.33%
North	4,049	37.07%
South	701	6.42%
Unknown / Multi-Regional	675	6.18%
Total	**10,922**	**100.00%**

As detailed in Table 2,In the information that was gathered, Central and Northern Italy account for **87.40%** of API-harvestable water metadata records in the country (50.33% and 37.07%, respectively). In stark contrast, the entire Southern region accounts for **6.42%** (701 records) [71].

Table 3: Metadata Extracted per Italian Region via Automated Harvesting [71]

Region / Territory	**Record Count**	**Percentage (%)**
Toscana	5,218	47.78%
Emilia-Romagna	1,633	14.95%
Trentino-Alto Adige / Südtirol	741	6.78%
Place-specific / Localized	675	6.18%
Veneto	533	4.88%
Piemonte	508	4.65%
Lombardia	482	4.41%
Basilicata	186	1.70%
Sardegna	167	1.53%
Campania	131	1.20%
Umbria	122	1.12%
Lazio	115	1.05%
Liguria	104	0.95%
Puglia	88	0.81%
Calabria	78	0.71%
Marche	42	0.38%
Sicilia	33	0.30%
Valle d'Aosta	28	0.26%
Friuli-Venezia Giulia	20	0.18%
Molise	10	0.09%
Abruzzo	8	0.07%
Total	**10,922**	**100.00%**

At the regional level (Table 3), Toscana alone generates 47.78% (5,218 records) of all metadata harvested across Italy, driven primarily by the extensive geoportal endpoints of the

LaMMA consortium (geoportale.lamma.rete.toscana.it). Emilia-Romagna represents the second largest provider with 14.95% (1,633 records). Combined, these two regions supply 62.73% of the entire national water metadata harvested pool.

Conversely, large southern regions with critical water supply infrastructure—such as Puglia (88 records, 0.81%), Calabria (78 records, 0.71%), Sicilia (33 records, 0.30%), and Abruzzo (8 records, 0.07%)—exhibit a severe deficit in API-accessible open data. This disparity demonstrates that open data availability is not correlated with infrastructure scale or water vulnerability, but rather with regional administrative commitment to open data standards.

Format Distribution and System Interoperability

Harvested metadata records were categorized by file extension and service format to evaluate their technical suitability for multimodal data integration.

Table 4: Distribution of Harvested Metadata Records by Resource Format / Type

Format / Extension Category	Resource Count	Percentage (%)
PNG Image (png)	4,597	42.09%
Shapefile Vector (shp)	2,571	23.54%
Web Service (srvc)	1,561	14.29%
Map Service / Preview (map)	1,457	13.34%
XML Metadata (xml)	924	8.46%
HTML Web Page (html)	906	8.30%
Comma-Separated Values (csv)	788	7.21%
Document File (doc)	773	7.08%
OP / DATPRO Format (op / datpro)	730	6.68%
ZIP Archive (zip)	629	5.76%
GeoJSON Vector (geojson)	285	2.61%
JSON Data (json)	253	2.32%
TIFF Raster (tiff)	224	2.05%
KML Vector (kml)	166	1.52%
Web Map Service (wms)	145	1.33%
Geography Markup Language (gml)	122	1.12%

The breakdown in Table 4 reveals significant structural characteristics:

- **Dominance of Cartographic Previews:** Static image files (png, 42.09%) and map previews (map, 13.34%) represent the largest format category. These resources typically correspond to rendered map tiles or static GIS previews rather than queryable, raw vector networks.
- **Core Spatial Vector Infrastructure:** Standard GIS vector formats—principally Shapefiles (shp, 23.54%), GeoJSON (geojson, 2.61%), KML (kml, 1.52%), and GML (gml, 1.12%)—total over 3,100 resources. These represent the primary structural assets for modeling pipeline assets, node topology, and relevant conextual information about the water system.
- **Live Web Services:** OGC-compliant web services (srvc, 14.29%; wms, 1.33%; wfs, 0.35%) provide live, streaming access to geospatial layers, facilitating automated integration without manual file downloads.

- **Tabular and Structured Data:** Machine-readable tabular datasets (`csv`, 7.21%; `json`, 2.32%; `xlsx`/`xls`, 1.27%) represent secondary resources, containing administrative, tariff, and water quality indicator series.

Temporal Metadata Void and Cadence Bottlenecks

A critical finding of the automated harvesting process concerns the extreme scarcity of standardized temporal metadata (Table 5).

Table 5: Metadata Extracted per Update Cadence Category

Update Cadence Attribute	Harvested Count	Percentage (%)
Unspecified / Missing (`<NA>`)	10,764	98.55%
Annual (`annual`)	88	0.81%
Unknown (`unknown`)	20	0.18%
Never Updated (`never`)	16	0.15%
Irregular (`irreg`)	9	0.08%
Daily (`daily`)	7	0.06%
Monthly (`monthly`)	7	0.06%
Continuous (`cont`)	6	0.05%
Bimonthly (`bimonthly`)	3	0.03%
Biweekly (`biweekly`)	2	0.02%
Total	**10,922**	**100.00%**

An overwhelming **98.55%** of all harvested metadata records (10,764 out of 10,922) lack any explicit update cadence definition (`<NA>`). Only **88 records (0.81%)** are explicitly tagged as annually updated, while near-real-time updates—such as daily (0.06%) or continuous (0.05%) feeds—represent negligible fractions.

This metadata void introduces a severe technical bottleneck for automated pipeline failure prediction systems. Without explicit cadence tags, data ingestion pipelines cannot programmatically determine whether a pipeline network dataset reflects live operational monitoring or an abandoned historical survey.

Host Domain Infrastructure Analysis

Evaluating the primary domain endpoints (Table 6) illustrates the infrastructure backbone supporting Italian open water data.

The national open data gateway (`dati.gov.it`) serves as the single largest metadata host, accounting for **38.62%** (4,218 records) of the harvested pool [71]. The LaMMA regional domain pair in Tuscany supplies 27.74% (3,030 combined records), followed by regional GIS servers in Emilia-Romagna, Trentino, Piemonte, and the Po River Basin Authority (`adbpo.it`).

This hosting distribution demonstrates that while `dati.gov.it` effectively aggregates metadata across municipalities and national agencies, direct API harvesting of regional GIS endpoints is essential to discover high-resolution spatial vector layers and live web services necessary for multimodal water network modeling.

Table 6: Top Host Domains Supplying Harvested Water Infrastructure Metadata

Hosting Domain / Portal Endpoint	Harvested Count	Percentage (%)
`dati.gov.it` (National Open Data Portal)	4,218	38.62%
`geoportale.lamma.rete.toscana.it`	1,515	13.87%
`lamma.rete.toscana.it`	1,515	13.87%
`mappegis.regione.emilia-romagna.it`	266	2.44%
`servizigis.regione.emilia-romagna.it`	249	2.28%
`webgis.adbpo.it` (Po River Basin Authority)	178	1.63%
`servizimoka.regione.emilia-romagna.it`	133	1.22%
`ambiente.regione.emilia-romagna.it`	113	1.03%
`siatservices.provincia.tn.it`	113	1.03%
`webgis.arpa.piemonte.it`	104	0.95%

3.2 Data Selection

Although automated API harvesting yielded 10,922 metadata records, a systematic quality audit revealed that the raw pool was too noisy, unstandardized, and geographically imbalanced to be directly ingested into a failure prediction pipeline. Over 98.55% of records lacked standardized temporal update cadences, and 42.09% consisted of static renders (`png`), CAD files (`dxf`), or administrative text rather than machine-readable telemetry.To establish a methodologically sound benchmark, data acquisition was initially delimited to the Region of Tuscany, leveraging its exceptionally rich spatial repository, high-resolution elevation models, and comprehensive hydrogeological data. To test and validate the proposed methodology, the study area was subsequently refined to the Province of Florence. This provincial scope eliminates severe spatial sparsity, maximizes signal-to-noise ratio, and provides a controlled, high-utility proof-of-concept incorporating satellite ground motion, soil vulnerability. Crucially, this provincial framework aims to serve as a reproducible baseline that can be seamlessly scaled or adapted across other Italian provinces and regions.

- **Geospatial Administrative Boundaries**
  - **Description:** Official statistical administrative boundaries provided by ISTAT at national, regional, provincial, and municipal levels, including census tract divisions (Basi Territoriali).
  - **Target Layers:** Italian municipal boundaries (Comuni) and detailed provincial boundaries for Florence (Firenze).
  - **Format & Coverage:** Polygon vector shapefiles; National and Regional (Toscana) coverage [74, 75].
- **Landslides and Flood Risks (ISPRA IdroGEO)**
  - **Description:** National open data platform managed by ISPRA providing hydrogeological hazard, risk indicators, and historical event tracking.
  - **Target Layers:** Italian Landslide Inventory (IFFI), national flood hazard maps, municipal risk indicators, and specific Tuscan landslide event layers.
  - **Format & Coverage:** Vector spatial points/polygons and tabular indicators; Italy / Region of Tuscany [76].

- **Soil Defense and Disaster Interventions (ReNDiS)**
  - **Description:** National repository monitored by ISPRA listing public engineering works and financial allocations for hydrogeological risk mitigation.
  - **Target Layers:** Georeferenced intervention points with financial and metadata attributes.
  - **Format & Coverage:** Georeferenced points and spatial database; Italy / Tuscany [77].
- **Terrain and Surface Digital Models (Regione Toscana)**
  - **Description:** High-resolution elevation models hosted on the Geoscopio Cartoteca and Tuscany Open Data Portal.
  - **Target Layers:** Tuscan DTM (Orogrfico, Idrologico, Esposizione Idrologico, Slope Idrologico), alongside 2023 DTM and DSM high-resolution rasters for Florence.
  - **Format & Coverage:** Raster grid layers (GeoTIFF / ASCII Grid); Region of Tuscany and Florence Municipality [78, 79].
- **Seismic Catalogs and Site Intensity (INGV)**
  - **Description:** Official seismic reference databases compiled by the Istituto Nazionale di Geofisica e Vulcanologia (INGV).
  - **Target Layers:** Parametric Catalog of Italian Earthquakes (CPTI15) and Macroseismic Italian Database (DBMI15).
  - **Format & Coverage:** Tabular / Spatial Point Datasets; Italy [80].
- **Hydro-Meteorological Network and Rainfall History (SIR Toscana)**
  - **Description:** Real-time and historical hydro-meteorological monitoring station measurements managed by the Hydrological Service of Tuscany (SIR).
  - **Target Layers:** Station series for Florence and Pisa provinces (rainfall mm, days of rain, extreme precipitation events > threshold mm), daily temperature, and wind speed (Anemometria/Termometria).
  - **Format & Coverage:** Tabular hydro-meteorological records; Firenze Province [81].
- **Pedology and Soil Capacity (DBPed Toscana)**
  - **Description:** Tuscan regional pedological database detailing soil system classifications, capability, and land utilization.
  - **Target Layers:** DBPed soil type classifications and soil usage capacity maps.
  - **Format & Coverage:** Vector GIS shapefiles; Region of Tuscany [82].
- **Socio-Economic and Fragility Indicators (ISTAT / MEF)**
  - **Description:** Socio-economic datasets from ISTAT Data Browser and the Ministry of Economy and Finance (MEF).
  - **Target Layers:** Taxpayer counts and total income by brackets (IRPEF), Composite Fragility Index (Indice composito di fragilità), primary income categories, and municipal water consumption.

  - **Format & Coverage:** Tabular statistical aggregates; Municipal level (Comuni) [83, 84, 85].

- **Population Census (ISTAT)**
  - **Description:** Historical time series of resident population census from ISTAT's permanent census portal.
  - **Target Layers:** Resident population aggregated at the municipal scale.
  - **Format & Coverage:** Tabular; Italian Municipalities [86].

- **Hydrography and Water Distribution System (Dati.gov.it)**
  - **Description:** Detailed open datasets on Tuscan surface water bodies, culverts, water utility production, and public drinking points.
  - **Target Layers:** Water bodies (Specchi d'acqua), rivers (Corso d'acqua), production metrics for Anconella and Mantignano plants, public water fountains (Fontanelli High Quality), Bilancino reservoir levels, and culverted streams (Tombinati).
  - **Format & Coverage:** Vector network layers and tabular operational data; Florence Province and Tuscany Region [87, 88, 89, 90, 91, 92, 93, 94].

- **Transport Infrastructure and Built Environment (OSM / CTR)**
  - **Description:** Spatial infrastructure layers extracted via OpenStreetMap (OSM QGIS tools) and Regional Technical Maps (CTR).
  - **Target Layers:** Complete street networks (highways) and building footprint vectors.
  - **Format & Coverage:** Vector lines and polygons; Florence Province / Region of Tuscany [95].

- **Commuting Mobility Matrix (ISTAT)**
  - **Description:** Inter-municipal commuting matrix (Matrice del Pendolarismo) based on the 2021 Census.
  - **Target Layers:** Origin-destination matrices for work and study movements.
  - **Format & Coverage:** Matrix tabular data; Italian Municipalities [96].

- **Satellite Soil Displacement Data (Sentinel-1 InSAR)**
  - **Description:** Ground movement radar interferometry measurements derived from Sentinel-1 satellite observations, accessible via ArcGIS REST Server.
  - **Target Layers:** Continuous ground deformation and line-of-sight displacement rate points.
  - **Format & Coverage:** Spatial GIS web layers (ArcGIS REST); Northern Apennines River Basin Authority [97, 98, 99].

3.3 Data Transformation and cleaning

To construct a Spatial Dataset, the raw, heterogeneous datasets collected from European, national, and regional repositories must undergo systematic data transformation and spatial cleaning. In raw form, the acquired datasets exhibit conflicting spatial representations (vector points, linear networks, complex polygons, and sparse station coordinates), mismatched coordinate reference systems, and incompatible attributes. Direct tabular merging of categorical strings, spatial densities, and variables without spatial standardization introduces data duplication, topological overlap, and grid matrix distortion.

To resolve these spatial discrepancies, every data layer is transformed and downsampled into a uniform base spatial index: the `h310grid`, constructed using Uber's H3 Spatial Index at Resolution 10. The transformation strategy defines rigorous spatial operations and statistical aggregation rules tailored to each geometry type, establishing a perfect 1-to-1 feature matrix across all grid cells.

3.3.1 Geometry-Specific Transformation Principles

The data transformation protocol classifies input features into four fundamental geometric archetypes, applying specialized transformation rules to collapse multidimensional spatial features into single, standardized cell attributes:

1. **Point Layers (Infrastructure, Interventions, and Events):** Point features representing physical assets, fountains, intervention sites, landslides, and seismic events are transformed using spatial joins (`gpd.sjoin`). For localized assets and events across the cell and its immediate 1-ring hexagonal neighborhood. Variable-specific rules include:

    - Summation of financial allocation columns (e.g., total intervention cost `importo_lo`) and extraction of dominant intervention classes (`tipo_disse`).
    - Exact occurrence counts per hexagon for active public water fountains (`pubacq_fontanello`) and regional landslide inventory points (`frane_piff_toscana_opendata`).
    - Maximum elevation extraction (`ele`) for topographic peak coordinates (`_peaks`).
    - Nearest-neighbor distance mapping for municipal place names and localities (`_places`).
    - Event counts coupled with maximum and mean moment magnitude (`MwDef`) calculations for localized seismic events (`seismic_points_utm32n`).

2. **Point Layers (Sparse Environmental Monitoring Networks):** Weather and hydrological monitoring stations (thermometers, pluviometers, hydrometers, and anemometers) are irregularly scattered across the territory, making direct spatial joins unfeasible as they would leave the vast majority of grid cells empty. These sparse observations are transformed into continuous spatial surfaces. Temperature, precipitation, and wind velocity metrics are generated via spatial interpolation techniques (Inverse Distance Weighting or Kriging) aggregating continuous monthly values directly into hexagon centroids. For river stage and hydrometric stations, proximity metrics are computed by calculating the exact Euclidean distance from each cell centroid to the nearest active monitoring station.

3. **Line Layers (Networks and Waterways):** Linear geometries cannot be directly assigned to polygon grid cells without geometric conversion into continuous spatial metrics. Linear features are transformed into two primary representations:

- *Network Density:* Road network vectors (`_road`) are intersected with grid geometries using spatial overlay operations (`gpd.overlay`). Total line length inside each hexagon is computed and divided by the cell surface area to yield localized road density (m/m^2).
- *Proximity and Classification:* Waterway network lines (`waterwaysL`) are transformed by computing the exact Euclidean distance from each cell centroid to the nearest line segment using `gpd.GeoSeries.distance`, alongside extracting the dominant watercourse classification passing through the cell.

4. **Polygon Layers (Soil, Hazards, Land Use, and Administrative Polygons):** Overlapping polygon layers present the risk of multi-category fragmentation. To collapse overlapping multi-polygon attributes into a single row per cell, two primary aggregation rules are applied:

   - *Majority Area Rule:* Intersecting polygon layers—such as root zone depth (`profondita_radici`), surface stoniness (`pietrosita_superficiale_`), hydraulic conductivity (`ksat_30`, `ksat_150`), soil erosion (`erosione_potenziale_tha`), land capability, and landscape units (`unita_di_paesaggio`)—are fragmented against the grid boundaries using `gpd.overlay`. The surface area of each fragmented polygon within the cell is calculated, and the attribute covering the maximum surface area is assigned as the dominant cell class.
   - *Ordinal Max Priority Rule:* For environmental hazard maps (e.g., ISPRA hydraulic risk `aggr_mosaicatura_ispra_2020_pericolosita_idraulica_firenze` and PAI landslide hazard layers `mosaicatura_ispra_2024_pericolosita_frana_pai`), categorical classifications are converted into an ordinal numerical scale (0 = None to 5 = Very High). For each hexagon, all intersecting hazard polygons are evaluated, and the cell is assigned the maximum hazard score present (max(values)), ensuring that localized severe risks are preserved rather than averaged out.
   - *Centroid Structural Assignment:* Large administrative boundaries (municipal census tracts, income distributions, and water balance totals) are mapped using spatial joins on cell centroids (`gpd.sjoin(predicate='within')`). Attributes from the encompassing administrative polygon are transferred directly to all contained hexagon centroids, establishing localized socio-economic baselines and network water loss metrics (Water Loss = ACQ_IMM − ACQ_EROG).

3.3.2 Pipeline Integration Methodology and Execution Architecture

The execution of these data transformation protocols is orchestrated through a modular, multi-stage Python pipeline. Rather than executing a monolithic, memory-intensive process, the integration workflow is decomposed into six sequential execution scripts. Each script processes a specific data archetype, persisting intermediate GeoPackage artifacts to ensure pipeline modularity and auditability.

1. **Step 1: Foundational Grid Preparation:** Ingests the master `h310grid` spatial template layer, initializing the empty spatio-temporal matrix and establishing spatial indexing.

2. **Step 2: Point Feature Integration (`1_integrate_points_sjoin.py`):** Loads the master grid and executes vector spatial joins (`gpd.sjoin`) across all point layers. Data points falling within each hexagon (or its immediate 1-ring neighbor context) are grouped by cell index (`h3_index`) to produce the intermediate `master_grid_with_points.gpkg`.

3. **Step 3: Linear Feature Integration (`2_integrate_lines.py`):** Computes linear spatial metrics by combining intersection overlays (`gpd.overlay`) and centroid distance calculations (`gpd.GeoSeries.distance`) outputting `master_grid_with_lines.gpkg`.

4. **Step 4: Complex Polygon Integration (`3_integrate_polygons_overlay.py`):** Processes polygons, intersected with grid boundaries, calculating fragment surface areas. The script enforces the *Majority Area Rule* and the *Ordinal Max Priority Rule*outputting `master_grid_with_polygons.gpkg`.

5. **Step 5: Administrative Centroid Integration (`4_integrate_polygons_centroid.py`):** Maps broad municipal-level administrative, demographic, and economic indicators. Using a spatial point-in-polygon predicate (`predicate='within'`), the script evaluates each hexagon centroid, transferring municipal population census counts, and utility water loss figures directly to contained cells to produce `master_grid_with_communal.gpkg`.

6. **Step 6: Monitoring Station Interpolation and Distance Mapping (`5_integrate_stations.p`** Processes time-series data from sparse environmental monitoring stations. Surface meteorology variables (temperature, precipitation, wind speed) are aggregated and spatially interpolated across the grid using IDW and Kriging models, while hydrometric river stage stations are processed via nearest-neighbor spatial joins (`gpd.sjoin_nearest`) to compute proximity distances, producing `master_grid_with_stations.gpkg`.

7. **Step 7: Final Data Cube Synthesis and Consolidation (`6_finalize_datacube.py`):** Executes the final aggregation and cleaning pass. The script merges pre-computed Digital Terrain Model (DTM) zonal statistics directly via native index joins, concatenates all intermediate feature tables into a consolidated wide GeoDataFrame, performs missing value imputation, validates schema data types, and exports the final, fully integrated `FINAL_DATA_CUBE.gpkg`.

Through this modular architecture, raw multi-source geographical and statistical datasets are transformed into a clean, standardized, and fully aligned spatial data cube ready for predictive failure modeling.

3.4 Data summary statistics

Prior to evaluating spatial patterns, aspatial central tendency and distribution metrics were analyzed to assess data completeness, skewness, and category dominance across the study domain.

3.4.1 Ordinal and Discrete Quantitative Metrics

Table 7 presents summary statistics for integer-based variables, including binary presence indicators and discrete count data.

Key observations regarding integer variables include:

- **Binary Soil Indicators:** Both `descending_soil_presence` and `ascending_soil_presence` exhibit a mode and 75th percentile of 0, with low mean values (≈ 0.015 and 0.014, respectively). This reflects extreme spatial sparsity, confirming that specific soil horizon conditions exist in only approximately 1.4%–1.5% of active grid cells across the domain.

Table 7: Summary Statistics (Integers)

Variable	Min	25%	50%	75%	Max	Mean	Mode
descending_soil_presence	0	0	0	0	1.0e+00	1.5e-02	0
ascending_soil_presence	0	0	0	0	1.0e+00	1.4e-02	0
census_pop	1.1e+03	4.8e+03	1.3e+04	1.8e+04	3.7e+05	2.4e+04	4.6e+03
inflow_total	2.0e+02	1.3e+03	3.9e+03	6.1e+03	1.8e+05	9.7e+03	1.3e+03

- **Demographic and Flow Distributions:** Population density (`census_pop`) and inflow volumes (`inflow_total`) demonstrate pronounced positive (rightward) skewness. For `census_pop`, the mean (2.4×10^4) significantly exceeds the median (1.3×10^4), driven by dense urban clusters reaching maximum concentrations of 3.7×10^5. Similarly, `inflow_total` displays a median of 3.9×10^3 compared to a mean of 9.7×10^3, indicating highly concentrated commuter inflow channels.

3.4.2 Categorical Attributes and Completeness Analysis

Table 8 summarizes categorical variables, detailing missingness rates, category richness, and dominant modes.

Table 8: Summary Statistics (Categorical)

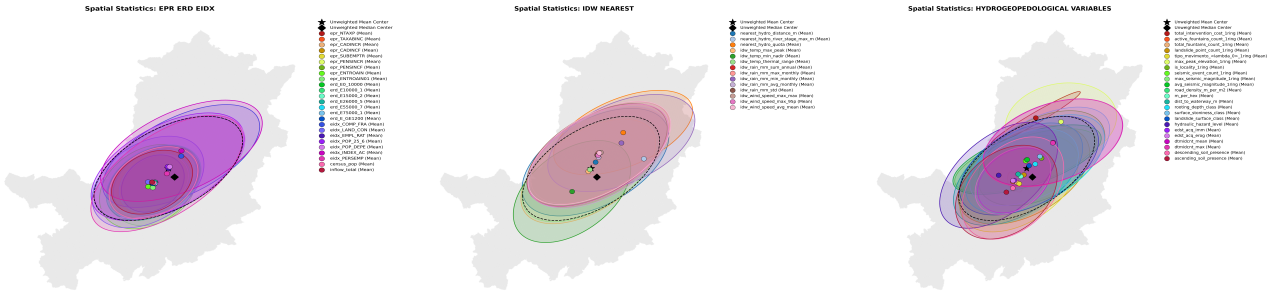
Variable	Missing Count	Missing %	Unique Categories	Most Frequent Value	Most Frequent Count	Most Frequent %
class_intervention_1ring	2.3e+05	1.0e+02	3.0e+00	Misto	3.5e+01	5.6e+01
locality_name_1ring	2.0e+05	8.8e+01	3.6e+03	Il Poggio	1.0e+02	3.9e-01
dominant_highway	1.0e+05	4.6e+01	2.9e+01	track	4.7e+04	3.9e+01
dominant_surface	1.7e+05	7.3e+01	3.5e+01	asphalt	2.9e+04	4.8e+01
dominant_tunnel	2.2e+05	9.9e+01	7.0e+00	yes	1.4e+03	8.2e+01
dominant_bridge	2.2e+05	9.8e+01	9.0e+00	yes	3.7e+03	9.2e+01
usda_hydrologic_group	1.2e+01	5.3e-03	5.0e+00	C	9.5e+04	4.2e+01
pai_landslide_hazard_level	4.2e+04	1.9e+01	5.0e+00	Moderata P1	7.6e+04	4.2e+01
dominant_building_1	1.7e+05	7.7e+01	7.6e+01	yes	4.4e+04	8.5e+01
dominant_building_2	2.2e+05	9.6e+01	8.5e+01	yes	2.7e+03	3.3e+01
dominant_building_3	2.2e+05	9.9e+01	7.6e+01	yes	4.3e+02	1.8e+01

An evaluation of categorical metrics reveals key structural properties of the dataset:

- **Data Coverage and Completeness:** Environmental and physical variables show high spatial coverage. Specifically, `usda_hydrologic_group` is nearly fully populated with only a 0.0053% missing rate, where Group C (moderately high runoff potential) is the dominant class (42%). Similarly, `pai_landslide_hazard_level` has an 81% coverage rate, with "Moderata P1" representing the primary hazard level (42%).

- **Sparsity in Specialized Infrastructure:** Features related to localized infrastructure exhibit high missingness percentages (e.g., `dominant_tunnel` at 99%, `dominant_bridge` at 98%, and secondary/tertiary buildings `dominant_building_2/3` at $> 96\%$). This high missingness reflects genuine spatial absence across rural grid cells rather than data loss.

- **High-Cardinality vs. Dominant Modes:** Variables such as `locality_name_1ring` exhibit high cardinality ($3,600$ unique categories) with a low top-mode frequency (0.39%), reflecting localized administrative dispersion. Conversely, infrastructure attributes show strong modal concentration; for example, unpaved agricultural paths (`track`) constitute 39% of mapped road networks, while paved surfaces (`asphalt`) dominate surface types (48%).



(a) Socio-economic factors (b) Environmental factors (c) Hydrogeopedological factors

Figure 1: Central tendencies and spatial dispersion

To evaluate the central tendencies, spatial dispersion, and directional anisotropy of variables across the study region, Standard Deviational Ellipses (SDE) and Weighted Mean Centers were calculated across the H3 hexagon grid domain.

3.4.3 Spatial statistics: mathematical framework

For any continuous numeric variable w_i observed at spatial locations (x_i, y_i) corresponding to n grid cell centroids, the Weighted Mean Center $(\bar{X}_w, \bar{Y}_w)$ represents the spatial center of mass and is defined as:

$$\bar{X}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}, \quad \bar{Y}_w = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

The Standard Deviational Ellipse (SDE) summarizes three core spatial properties simultaneously:

1. **Centrality:** The weighted spatial location of the attribute distribution.
2. **Dispersion (1σ):** The standard deviation along two orthogonal axes (σ_x and σ_y), defining the major axis $a = \max(\sigma_x, \sigma_y)$ and minor axis $b = \min(\sigma_x, \sigma_y)$.
3. **Orientation (θ):** The rotation angle relative to True North, determined by:

$$\tan\theta = \frac{\left(\sum \tilde{x}_i^2 - \sum \tilde{y}_i^2\right) + \sqrt{\left(\sum \tilde{x}_i^2 - \sum \tilde{y}_i^2\right)^2 + 4\left(\sum \tilde{x}_i \tilde{y}_i\right)^2}}{2\sum \tilde{x}_i \tilde{y}_i} \tag{3}$$

where $\tilde{x}_i = x_i - \bar{X}_w$ and $\tilde{y}_i = y_i - \bar{Y}_w$. An unweighted geometric baseline (weights = 1.0) provides the spatial benchmark representing the domain's boundary shape and geometric centroid.

3.4.4 Spatial Comparative Results and Analysis

Comparative visualization across the thematic variable groups highlights distinct spatial behaviors relative to the unweighted baseline ($\star$ Unweighted Mean Center, $\diamondsuit$ Unweighted Median Center, and dashed black SDE ellipse).

Socio-Economic and Demographic Factors

In Figure 1 in sub-plot 1a,analysis of socio-economic indicators (including income levels `TAXABINC`, `CADINCR`, dependency ratios, and population density `POP_25_6`) reveals the following spatial characteristics:

- **Central Tendency Shift:** The weighted mean centers are closely grouped near the geometric unweighted mean center, with a subtle North-Northeast displacement for indicators such as land consumption (`eidx_LAND_CON`) and composite population indices (`eidx_INDEX_AC`). This points to a concentration of human and economic activity in the central-northern sector of the domain.
- **Dispersion and Alignment:** The rotation angles (θ) and axis lengths closely mirror the unweighted geometric ellipse of the overall study boundary. The large ellipse footprint demonstrates that socio-economic dynamics operate on a broad, macro-regional scale rather than being confined to isolated pockets.

Environmental and Climatic Factors

In Figure 1 in sub-plot 1b, variables derived from inverse distance weighting (IDW) interpolation—comprising temperature extremes (`idw_temp_max_peak`, `idw_temp_min_nadir`), precipitation metrics (`idw_rain_*`), and river proximity—demonstrate pronounced spatial gradients:

- **Directional Gradient Shifts:** Unlike socio-economic metrics, environmental mean centers exhibit substantial spatial divergence. Peak temperature maximums (`idw_temp_max_peak`) shift significantly North-Northeast toward lower valley floors, whereas minimum temperature nadirs (`idw_temp_min_nadir`) shift South-Southwest toward higher elevation zones.
- **Boundary Extensions:** The minimum temperature ellipse extends southward beyond the standard unweighted boundary, reflecting continuous, regional-scale climatic trends governed by elevation gradients.

Hydrogeopedological and Terrain Variables

In 1 in 1c, physical terrain properties, soil characteristics, and geohazards (including landslide counts `landslide_*`, seismic events `seismic_*`, maximum elevation `max_peak_elevation`, and soil horizon presence) show the highest spatial heterogeneity:

- **Local Clustering:** Weighted mean centers are scattered across a broader central band rather than converging on a single point. For instance, `ascending_soil_presence` shifts markedly North, indicating specific soil horizon localization in northern sub-basins.
- **Anisotropy and Spatial Footprint:** Ellipses within this group exhibit variable orientations and eccentricities. High-elevation terrain variables align tightly along specific mountain ridges, while soil horizon and geohazard ellipses are noticeably smaller in area. This confirms that hydrogeopedological processes possess a highly localized footprint constrained by fine-scale topography, contrasting sharply with the domain-wide influence of socio-economic and climatic factors.

Filtering and Correlation Selection via Moran's I

To systematically evaluate spatial dependence across the extensive feature space, all numerical continuous, integer, and encoded attributes were first subjected to global Moran's I and local Moran's I testing. Global spatial autocorrelation was measured using row-standardized H3 grid spatial weight matrices (W), followed by pseudo-p-value significance testing.

To ensure methodological robustness, local Getis-Ord statistics (G_i^*) were computed alongside local Moran's I for all univariate attributes. While local Moran's I identifies spatial clustering and outliers, the Getis statistic specifically detects spatial concentration of high or low values, confirming that both test statistics converged on identical hotspot and coldspot structures across the domain.

Because dozens of variables were evaluated across the domain, only some attributes exhibiting statistically significant global spatial autocorrelation ($p < 0.05$, FDR-adjusted) and pronounced local clustering (high absolute Moran's I_i scores) are featured in the bivariate LISA reports. This filtering step ensures that the presented figures highlight key spatial dependencies, directional gradients, and localized hotspots while filtering out non-significant spatial noise.

The local bivariate Anselin Moran's $I_i^{(x,y)}$ for location i is defined as:

$$I_i^{(x,y)} = c_i \cdot z_{x,i} \sum_{j=1}^{n} w_{ij} z_{y,j} \tag{4}$$

where $z_{x,i} = \frac{x_i - \bar{x}}{\sigma_x}$ and $z_{y,j} = \frac{y_j - \bar{y}}{\sigma_y}$ represent standardized variable values, w_{ij} is the spatial weight matrix capturing neighbor topology across the H3 grid, and c_i is a scaling constant.

To prevent inflated Type I error rates arising from spatial autocorrelation across thousands of grid cells, pseudo-p-values derived from random spatial permutations were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure ($p < 0.05$). Significant grid locations were categorized into four local spatial quadrants:

1. **High-High (HH) Hotspots (Red):** High values of variable x surrounded by high values of variable y in neighboring cells.

2. **Low-Low (LL) Coldspots (Dark Blue):** Low values of variable x surrounded by low values of variable y in neighboring cells.

3. **High-Low (HL) Outliers (Light Orange):** High values of variable x surrounded by low values of variable y.

4. **Low-High (LH) Outliers (Light Blue):** Low values of variable x surrounded by high values of variable y.

5. **Non-Significant (ns) Areas (Light Grey):** Regions with no statistically significant spatial association after FDR correction ($p \geq 0.05$).

Socio-Economic and Environmental Integration Limitations Most socio-economic attributes showed statistically non-significant spatial correlations with environmental factors. Lacking pronounced spatial coupling, these variables cannot serve as interactive drivers for aggravated impact on already found failures, functioning instead as a macro-contextual backdrop. This decoupling mirrors known challenges in spatial data science, such as scale mismatches

between administrative census boundaries and continuous environmental grids, alongside disparate data-generating processes [100]. An exception emerged exclusively for population density (`census_pop`) and movement inflows (`inflow_total` representing *pendolarismo*), which displayed significant spatial synchronization and are the sole socio-economic metrics retained for further study.

Univariate results All significance maps presented have been adjusted for multiple testing using False Discovery Rate (FDR) control at $p < 0.05$.

The LISA maps for `census_pop` and `inflow_total` display near-identical spatial structure:

- **Central High-High (HH) Hotspot:** A concentrated urban core exhibiting statistically significant spatial clustering of dense population and elevated daily commuting inflow.
- **Peripheral Low-Low (LL) Coldspots:** Northern, north-eastern, and south-western rural hinterlands marked by sparse population density and low commuter movement.
- **Spatial Concordance:** The structural alignment between population distribution and daily inflow demonstrates that hydraulic demand peaks are geographically localized in the central core region.

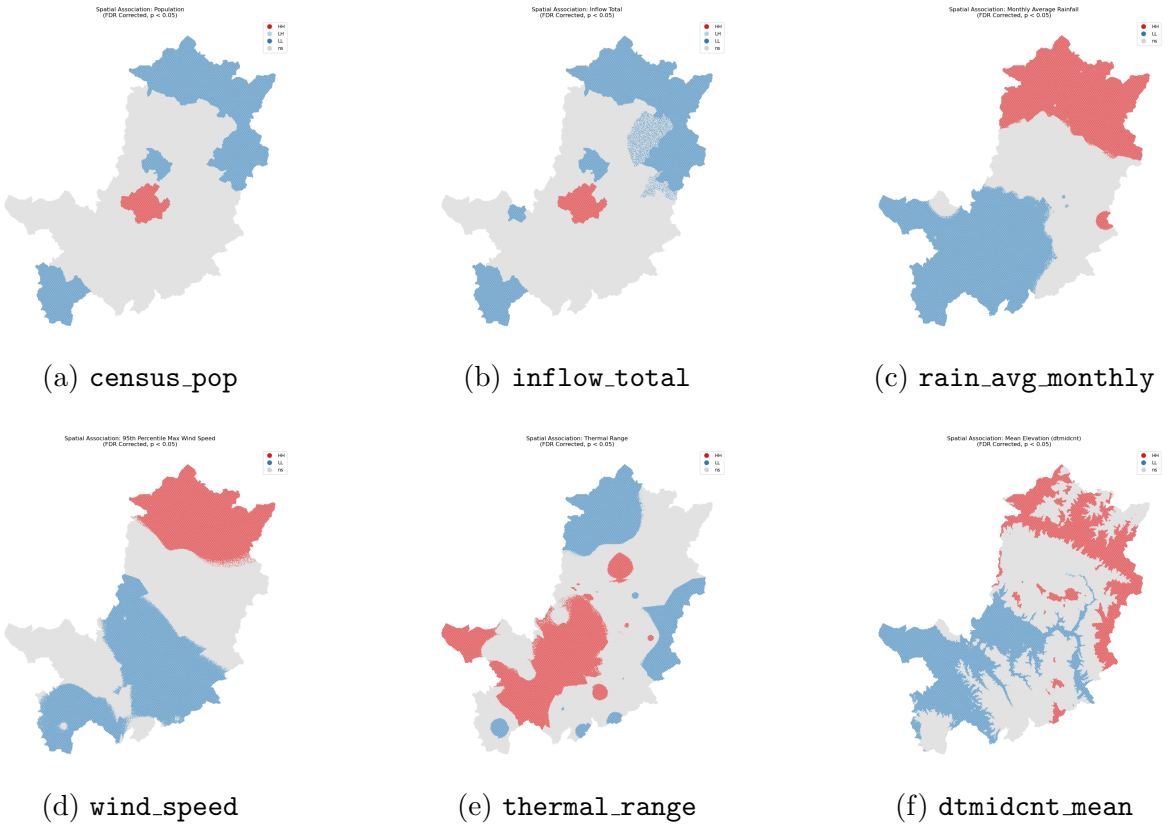
The LISA maps for environmental variables show higher activity in the central areas of the province and big groups of hot and cold spots:

- **Elevation (`dtmidcnt_mean`):** Displays an explicit topographical divide. High-High (HH) elevation clusters dominate the northern and eastern mountainous terrain, while Low-Low (LL) clusters cover the low-lying central-western river basin and coastal plain.
- **Monthly Average Rainfall (`idw_rain_mm_avg_monthly`):** Characterized by a strong north-south gradient. The northern mountainous zone forms a massive High-High (HH) precipitation cluster due to orographic uplift, whereas the southern region forms an extensive Low-Low (LL) drought-prone cluster.
- **Thermal Range (`idw_temp_thermal_range`):** Shows significant High-High (HH) hotspots in the central and south-western basins where continental microclimates induce wide diurnal temperature swings. Low-Low (LL) coldspots are observed in northern high-altitude zones.
- **Wind Speed (`idw_wind_speed_max`):** Features an intense High-High (HH) cluster across the northern mountain ridges (severe wind exposure) and a broad Low-Low (LL) region spanning the central-south lowlands.

Cross-Environmental Variable Interactions The physical integrity of water distribution networks is strongly affected by compound environmental stressors, some of which are hypothesis until we can find the causal links in the patents that can show us failure paterns. The spatial overlap of these variables generates three critical environmental interaction zones:

Table 9: Environmental Variable Interaction Matrix and Physical Impact on Water Networks

Interaction Zone	Spatial Overlap	Physical Infrastructure Stress
(High Elevation + High Rain + High Wind)	Northern Ridge (HH Elevation, HH Rain, HH Wind)	Orographic runoff causing slope instability, landslides, pipe shear stress, unseating of surface supply lines, and catchment turbidity spikes.
(Low Rain + High Thermal Range)	Central-South Basin (LL Rain, HH Thermal Range)	Soil moisture deficit, clay swelling/shrinkage cycles, ground motion leading to pipe bending, joint detachment, and accelerated burst rates.
(High Elevation Difference + Low Thermal Range)	Steep Terrain Boundaries	High hydrostatic static pressure head required for elevation change, increasing burst vulnerability during dynamic pressure surges.



(a) `census_pop` (b) `inflow_total` (c) `rain_avg_monthly`

(d) `wind_speed` (e) `thermal_range` (f) `dtmidcnt_mean`

Figure 2: Univariate Local Moran's I cluster maps for selected key variables across the H3 grid domain.

Bivariate results All significance maps presented have been adjusted for multiple testing using False Discovery Rate (FDR) control at $p < 0.05$.

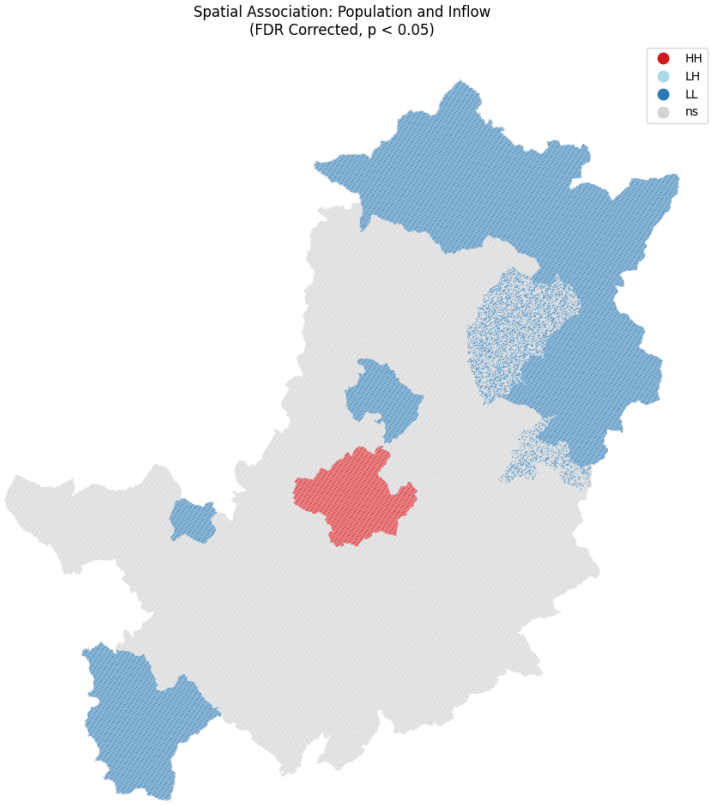


Figure 3: Demographic correlations census and inflows

Figure 3 illustrates the bivariate local spatial correlation between census population (`census_pop`) and total movement/traffic inflow (`inflow_total`).

- **Central Urban Core Hotspot (HH):** A prominent, contiguous High-High cluster (red) dominates the central portion of the territory. This indicates a strong spatial co-location where high residential population density directly coincides with high commuter/inflow hubs.
- **Peripheral Coldspots (LL):** Large Low-Low clusters (dark blue) are localized in the northern, eastern, and southwestern extremities of the domain. These areas correspond to sparsely populated rural peripheries that similarly experience low inflow intensity.
- **Spatial Homogeneity:** The near-total absence of spatial outliers (HL or LH) confirms that population distribution and functional movement inflows are tightly synchronized across the urban hierarchy.

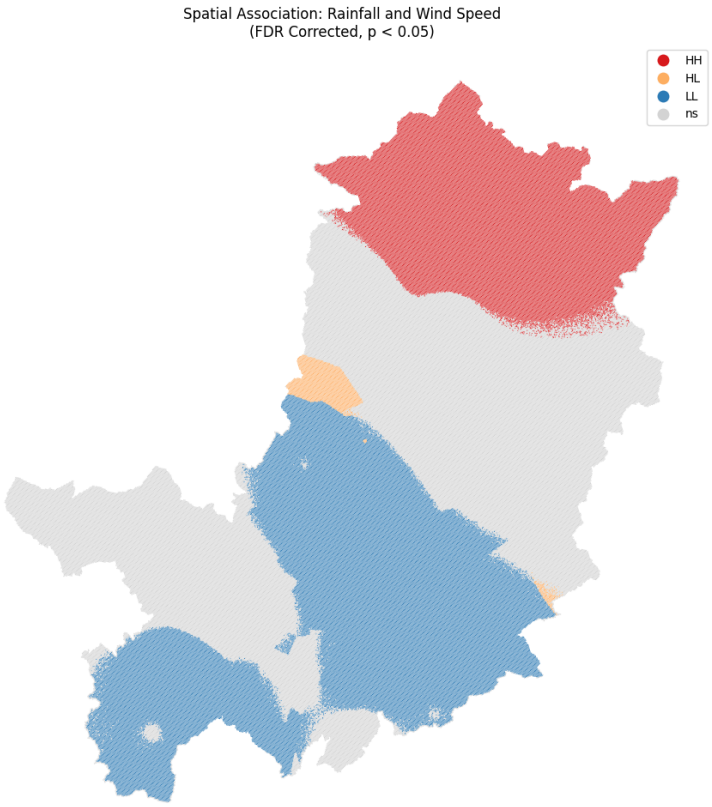


Figure 4: Environmental correlations rain and wind

Figure 4 maps the spatial coupling between average monthly precipitation (`idw_rain_mm_avg_monthly`) and extreme wind speed (`idw_wind_speed_max_95p`).

- **Northern High-Exposure Corridor (HH):** A distinct, extensive High-High hotspot (red) spans the northern alpine/mountainous rim of the region. This confirms a microclimatic regime where intense rainfall coincides with high wind exposure along exposed ridge lines.
- **Southern Basin Coldspot (LL):** Conversely, the central-southern and southwestern lowlands form a massive Low-Low coldspot (dark blue). These sheltered inland basins consistently experience both lower average precipitation and lower peak wind velocities.
- **Transitional Outliers (HL):** A small High-Low outlier patch (orange) occurs along the central boundary where wind velocities begin to decline ahead of rainfall drops, marking a physical transition zone.

Figure 5 evaluates the local relationship between annual thermal range (`idw_temp_thermal_range`) and mean elevation (`dtmidcnt_mean`).

- **Lowland High-Thermal Variance Outliers (HL):** A broad High-Low spatial cluster (orange) dominates the southwestern plateau and valley corridors. This highlights low-elevation geographical zones that experience high diurnal/seasonal thermal fluctuations due to valley thermal inversion and reduced moderating elevation effects.
- **High-Altitude Low-Thermal Variance Outliers (LH):** Extensive Low-High clusters (light blue) cover the high-elevation northern and eastern mountain belts. Here, higher

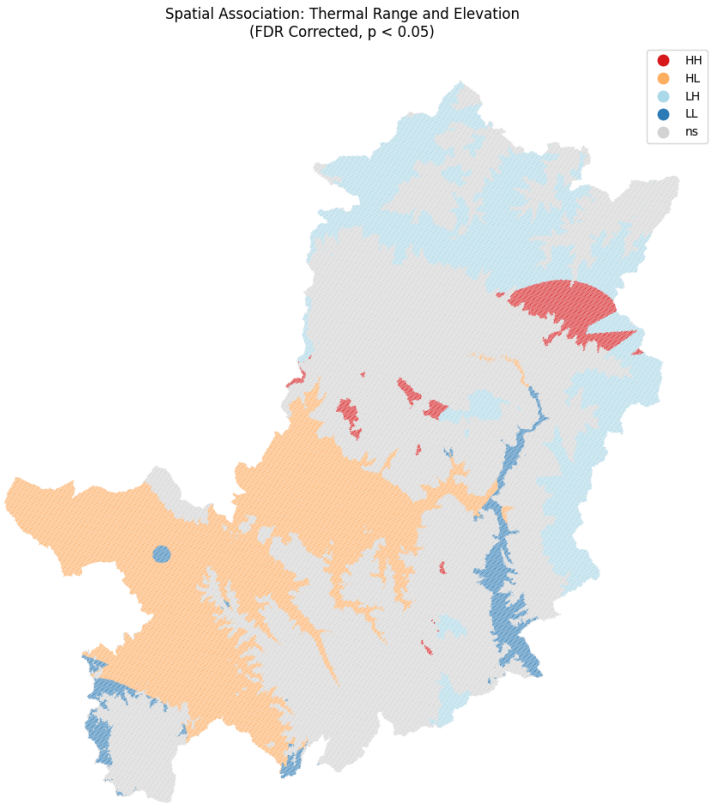


Figure 5: Geological correlations thermal ranges and elevation levels

mean elevation correlates with a narrower thermal range, reflecting cooler, highly buffered mountain climates.

- **Localized High-Altitude Hotspots (HH):** Secondary High-High hotspots (red) emerge along specific northeastern ridgelines, identifying localized high-altitude micro-climates that undergo extreme temperature swings.

Comparing the bivariate LISA results across domain areas underscores fundamental differences in spatial organization:

- **Socio-economic structures** display a strong monocentric spatial configuration (Figure 3), characterized by a single central HH core surrounded by peripheral LL zones.
- **Climatic interactions** follow a broad North-South regional gradient (Figure 4), dividing high-exposure northern mountains from sheltered southern plains.
- **Topographic-thermal interactions** exhibit high spatial complexity (Figure 5), dominated by inverse HL and LH spatial relationships driven by elevation gradients and local terrain geometry.

Crucially, mapping these spatial correlations and variable behaviors establishes a interconnected network of potential relationships across the dataset. This spatial network provides a structured framework for identifying and tracing how extreme, interconnected attribute values propagate across neighboring cells within the H3 grid domain.

Final Variable Selection The final selection of variables integrates key socio-economic, climatic, and geological dimensions that exhibit strong spatial coupling and localized clustering across the H3 grid domain. Socio-economic attributes like population (`census_pop`), movement inflows (`inflow_total`), and road density (`m_per_hex`) capture the monocentric urban core where residential density and commuting are tightly synchronized. Concurrently, environmental and climatic factors—such as maximum monthly rainfall (`idw_rain_mm_max_monthly`), peak wind speed (`idw_wind_speed_max_max`), and thermal range (`idw_temp_thermal_range`)—reflect distinct regional micro-climates and topographic gradients. Finally, terrain and hazard variables including maximum elevation (`dtmidcnt_max`), seismic magnitude (`avg_seismic_magnitude_1ring`), soil movement velocities (`avg_descending_soil_speed`, `avg_ascending_soil_speed`), landslide density (`landslide_point_count_1ring`), and hydraulic hazard (`hydraulic_hazard_level`) map critical physical vulnerabilities and structural stability risks across the territory's diverse altitudinal zones.

3.5 Patent data case

3.5.1 Source selection

To complement public administrative and geographical datasets with technological innovation insights, an exploratory investigation was conducted to identify relevant patents concerning pipeline integrity, leak detection, predictive maintenance, and water network analytics. Preliminary search queries were designed and benchmarked across four major intellectual property and patent discovery platforms:

- **The Lens** (`thelens.org`)
- **WIPO PATENTSCOPE** (`patentscope.wipo.int`)
- **EPO Espacenet** (`worldwide.espacenet.com`)
- **Google Patents** (`patents.google.com`)

Following an evaluation of search interface constraints, API throughput, and rate limiting policies, **The Lens** (`thelens.org`) was selected as the main source for patent metadata collection. The primary factor behind this choice was the platform's generous rate limits for searching and downloading dataset records, its support for structured API exports in JSON and CSV formats, and its generous academic access policies. Unlike other portals that impose restrictive web-scraping controls, captcha challenges, or tight record download ceilings, The Lens allowed for continuous, large-scale indexing of patent records necessary for identifying technological trends in water transport infrastructure monitoring.

3.5.2 Query design

The construction of the patent search query was a pivotal methodological step, designed to systematically harvest relevant patent documents from The Lens (`https://www.lens.org`) for subsequent causal link synthesis and technical solution mapping. The primary challenge

in retrieving patents related to utility-scale water distribution, conveyance, and environmental monitoring lies in the pervasive noise generated by consumer-facing and domestic water products. Key terms such as *"water"*, *"pipe"*, or *"conduit"* are heavily used in patents describing household appliances, kitchen fixtures, sanitization devices, and localized plumbing (e.g., faucets, showerheads, water carafes, or coffee makers). Left unconstrained, a broad keyword query retrieves thousands of irrelevant consumer-scale inventions, severely diluting the dataset's thematic integrity.

To overcome this, an iterative query engineering methodology was adopted. Custom keyword arrays, proximity operators, regular expressions with wildcards, and explicit International Patent Classification (IPC) and Cooperative Patent Classification (CPC) restrictions were combined to maximize signal-to-noise ratio. Throughout the development phase, iterative precision evaluations were conducted whenever a new boolean operator or exclusionary term was introduced. This calibration exercise ensured that candidate negative filters selectively eliminated undesired consumer or off-target industrial patents without accidentally pruning relevant infrastructure-level documents.

Iterative Filtering and Query Architecture The final query architecture, integrates four distinct functional layers designed to enforce domain specificity while capturing multi-faceted technical solutions:

1. **Classification Scoping (IPC/CPC):** Inclusions were targeted toward major conveyance and treatment classifications, specifically IPC classes `E03B*` (water supply systems), `E03F*` (sewerage), `F17D*` (pipe-line systems), and `C02F*` (treatment of water, waste water, and sewage), alongside CPC symbols `"E03B7/00"` and `"E03C1/02"`.

2. **Infrastructural & Domain Keywords:** Title and abstract fields were queried using wildcard stems and proximity expressions (e.g., `"water pipeline"`$\sim$5, `"water distribution"`$\sim$5, `"buried pipe"`$\sim$5, `"water grid"`, `aqueduct*`, `watercourse*`, `wastewater*`, and `conduit*`) to anchor the corpus within large-scale utility operations.

3. **Sensing, Environmental, & Stressor Keywords:** To capture inventions addressing structural monitoring, environmental impacts, and predictive maintenance, a secondary keyword block was joined via the `AND` operator. This block spans monitoring techniques (`telemetry`, `IoT`, `radar`, `fiber-optic*`, `drone*`, `satellite*`, `wireless`, `"anomaly detection"`$\sim$5), dynamic physical parameters (`pressure`, `stress`, `strain`, `vibration*`, `corrosion`, `cracking`, `leak*`, `"hydraulic gradient"`, `"flow rate"`$\sim$3), and environmental/seismic forces (`seismic*`, `subsidence`, `liquefaction`, `freeze-thaw`, `washout*`, `backfill*`, `soil`, `moisture`).

4. **Explicit Exclusionary Rules (`NOT` Clause):** To cut consumer noise and off-target industries, strict exclusions were imposed across titles, abstracts, and classification codes. The query explicitly rejected non-target fields such as oil/gas processing (`gas`, `oil`, `petroleum`, `molecular`), household fixtures (`faucet*`, `shower*`, `refrigerator*`, `sink*`, `carafe*`, `pitcher*`, `bottle*`, `flask*`, `straw*`, `countertop*`, `"beverage maker"`, `"garden hose"`, `"plumbing fixture"`, `spa`), and domestic plumbing IPC/CPC subclasses (`C02F1/003`, `C02F1/68`, `E03B9/00`, `E03C`, `E03D`, `E03B1/04`, `E03B11/02`).

Search Metadata and Metadata Filters Beyond thematic keywords and class codes, standardized bibliographic metadata parameters were applied to refine the search target. The

publication date was constrained to the range between January 1, 1950, and January 23, 2026. The search scope was restricted to four major global IP authorities: the United States (`US`), the European Patent Office (`EP`), Japan (`JP`), and the Republic of Korea (`KR`). Eligible document languages were set to English (`en`), Spanish (`es`), French (`fr`), Italian (`it`), and Portuguese (`pt`). Additionally, documents were grouped by simple patent family to avoid redundant count of equivalent filings, and restricted to granted patents to guarantee higher technical specificity.

Executing this exact query on Lens.org yielded a total primary dataset of $N = 4,779$ granted patent family documents.

Statistical Precision Validation To rigorously quantify the thematic precision of the final query prior to full-text extraction, a representative random sample was extracted for manual evaluation. The required sample size n was calculated using the standard statistical sampling formula for finite populations at a 95% confidence level ($Z = 1.96$) and a 5% margin of error ($e = 0.05$), assuming maximum population variance ($p = 0.5$):

$$n = \frac{\frac{Z^2 \cdot p \cdot (1-p)}{e^2}}{1 + \left(\frac{Z^2 \cdot p \cdot (1-p)}{e^2 \cdot N} \right)} \tag{5}$$

Substituting $N = 4,779$ into the formula yielded a required statistical sample size of $n = 356$ patents.

Each patent in the 356-document subsample was evaluated against explicit inclusion/exclusion taxonomy criteria:

- **Relevant (`1`):** Inventions covering core water infrastructure and conveyance (pipelines, aqueducts, sewage, utility distribution), treatment/management at utility scale, or structural/environmental monitoring factors (leaks, soil deformation, corrosion, seismic hazards, weather stressors).
- **Irrelevant (`0`):** Inventions dedicated strictly to domestic/micro-scale applications (household appliances, drinkware, localized plumbing fixtures) or off-target industrial processes (oil/gas extraction, laboratory molecular chemistry).

Out of the 356 evaluated patents, 296 documents met the strict relevance criteria (`1`), while 60 were identified as off-target (`0`). This yielded a validated precision rate of **83.15%** (with 16.85% false positives), confirming that the iterative filtering strategy successfully isolated utility-scale water infrastructure technology.

Full-Text Processing Pipeline Following precision verification, full-text extraction was performed across the complete set of fetched patents. This comprehensive full-text parsing was enabled with technical support from Erre Quadro, providing structured textual corpora (claims, descriptions, and technical specifications) necessary for automated causal link detection and technical solution analysis in subsequent thesis chapters, the final list of patents obtained yielded 17,987 patents since they were extracted from Erre Quadro's internal engine and not directly from The Lens.

3.5.3 Patent processing & summary statistics

To establish a structured corpus suitable for causal analysis, raw patent documents were retrieved and processed through a multi-stage filtering and segmentation pipeline. The complete

data lineage and row-level transformation metrics across all processing stages are detailed below.

Initial Data Ingestion and Normalization Because raw patent data were fetched as unstructured JSON files, an initial ingestion and normalization phase was required. Each document was parsed and transformed into a standardized record schema comprising four principal attributes: `patent_id`, `patent_title`, `text_type`, and `text`. The `text_type` field explicitly identifies whether a given paragraph originates from the legal claims section or the descriptive body of the specification. At this stage, text was organized at the paragraph level, yielding an initial normalized dataset of 288,125 rows, as detailed in Table 10.

Table 10: Data Processing and Initial Normalization Summary

Process	**Row Count**
Files Processed Successfully	17,987 / 17,987
Total Claim Rows Extracted	155,639
Total Description Paragraphs	132,486
Total Dataset Rows Created	**288,125**

Domain-Specific Relevance Filtering Following normalization, a thematic relevance filter was applied to remove off-topic content. Using a specialized ontology of domain key terms tailored to the water transportation sector (discussed in depth in the methodology chapter), paragraphs failing to match relevant concepts were pruned. As shown in Table 11, this step filtered out 43.9% of the raw corpus, retaining 161,658 highly targeted paragraphs.

Table 11: Patent Filtering on Domain Relevance Terms

Stage	**Row Count**	**Retention (%)**
Total Initial Rows	288,125	100.0%
Remaining After Filtering	161,658	56.1%
Filtered Out	**126,467**	**43.9%**

Causal Term Filtering To isolate text likely to yield explicit causal linkages, a third pipeline stage filtered the domain-relevant paragraphs against a curated vocabulary of causal indicators. Paragraphs containing no explicit causal mechanisms were discarded. Table 12 outlines the results of this step, leaving 62,009 paragraphs (38.4% of the domain-relevant subset) that hold potential causal chains for downstream extraction.

Paragraph-to-Sentence Segmentation Because specific causal statements typically manifest within discrete sentence boundaries rather than full paragraphs, the filtered paragraphs were segmented into individual sentences. During this row explosion process, metadata mapping was preserved by assigning both a `patent_id` and a `paragraph_id` to each sentence. This ensures fine-grained sentence-level analysis while maintaining complete contextual traceability back to the parent document section. As shown in Table 13, sentence tokenization expanded the dataset significantly, resulting in 277,675 sentence-level rows.

Table 12: Causal Terms Filtering Results

Stage	Row Count	Retention (%)
Total Initial Rows	161,658	100.0%
Remaining After Filtering	62,009	38.4%
Filtered Out	**99,649**	**61.6%**

Table 13: Paragraph to Sentence Segmentation (Row Explosion)

Stage	Row Count	Change (%)
Paragraph-Level Rows (Input)	62,009	100.0%
Sentence-Level Rows (Output)	277,675	447.8%
Net Sentence Expansion	**+215,666**	**+347.8%**

Final Dataset Summary Statistics The final structured corpus comprises 215,299 sentences extracted across 17,054 unique patents. As summarized in Table 14, each patent yields an average of 12.62 relevant sentences, demonstrating a high density of candidate causal statements per document in the final dataset.

Table 14: Final Patent Dataset Summary Statistics

Metric	Value
Total Sentence Rows	215,299
Total Unique Paragraphs	48,674
Total Unique Patents	17,054
Mean Patent Appearances	12.62

3.6 Ontology and key term lists

To capture and structure causal relations within the water transportation and pipeline engineering domain, a multi-phase ontological framework was designed. This framework establishes the semantic vocabulary required to extract physical assets, operational threats, environmental exposures, and causal mechanics from unstructured patent texts. The methodology comprises two main phases: (1) multi-source domain lexicon gathering, followed by Large Language Model (LLM) enrichment, and (2) programmatic Semantic Web Knowledge Graph construction via RDFLib and Natural Language Processing (NLP).

3.6.1 Phase 1: Terminology Extraction and Domain Lexicon Mining

The lexicon forms the foundational layer for text filtering and causality extraction. To avoid bias and ensure comprehensive coverage, terms were systematically harvested across four distinct operational categories and enriched through LLMs.

Taxonomic Categories of the Domain Lexicon The extracted vocabulary is organized into four complementary functional classes:

1. **Technical Terms:** Concrete physical entities, assets, components, and structural mechanics within the pipeline and water transport network (e.g., *pressure relief valve, pipeline casing, pump, flange*).

2. **Attribute Terms (Environmental Exposure Ontology):** Exogenous spatial, environmental, and socio-economic variables that act upon the infrastructure or describe its surroundings (e.g., *hydrologic soil group, peak ground acceleration, building density, land use*).

3. **Causal Terms (Causal Relation Lexicon):** Syntactic cue phrases, verb predicates, and relational anchors that signal direct or indirect causality in text (e.g., *resulting from, leads to, attributed to, triggers*).

4. **Failure Terms (Failure Mode Repository):** Specific degradation mechanisms, structural defects, and threat categories that compromise pipeline integrity (e.g., *stress corrosion cracking, fatigue failure, burst, internal erosion*).

Multi-Repository Information Retrieval Initial seed terms were mined from standard domain repositories and authoritative standards:

- **Failure Terms Repositories:** Broad failure vocabulary was retrieved from Wikipedia category structures (*Mechanical failure, Structural failure, Pipeline integrity*). Technical standards were integrated from NACE/AMPP corrosion glossaries (e.g., *pitting, pitting factor, coating disbondment*) and European/American pipeline integrity standards including EN 1594, ISO 13623, and API 1160.

- **Causal Cues and NLP Repositories:** Causal predicates were built using standard text mining libraries and NLP phrase banks, categorized into direct causation, conditional/contributing factors, correlation, and attribution.

- **Environmental Variables Repositories:** Spatial and environmental risk variables were structured using Wikipedia environmental hazard taxonomies (*Hydrological hazards, Mass wasting, Soil erosion*) and formal OBO ontologies such as the **Environmental Ontology (ENVO)** and the Environmental Exposure Ontology (ExO).

LLM-Based Terminology Enrichment To expand the vocabulary beyond rigid dictionary terms, seed lists were processed through LLMs (specifically GPT-4 and Gemini 3). The models were prompted to generate domain-specific synonyms, technical variations, morphological inflections, and contextual paraphrases while preventing out-of-domain semantic drift.

3.6.2 Phase 2: Programmatic Knowledge Graph Generation and RDF Encoding

Following lexicon extraction, the terminology was compiled into a formal Semantic Web Knowledge Graph utilizing custom Python automation, `rdflib`, and `spaCy` for syntactic normalisation.

Architecture and Class Hierarchies The ontology is declared under the base URI namespace `http://www.semanticweb.org/thesis/pipeline-risk#` (`ex:`). The Knowledge Graph integrates W3C standard vocabularies including SKOS (Simple Knowledge Organization System), OWL (Web Ontology Language), RDF, and RDFS.

As illustrated in Figure 6, the graph is anchored by four primary SKOS Concept Schemes representing the four pillars of the domain:

1. `ex:TechnicalComponentScheme`
2. `ex:FailureModeScheme`
3. `ex:CausalRelationalScheme`
4. `ex:EnvironmentalSocialRiskScheme`

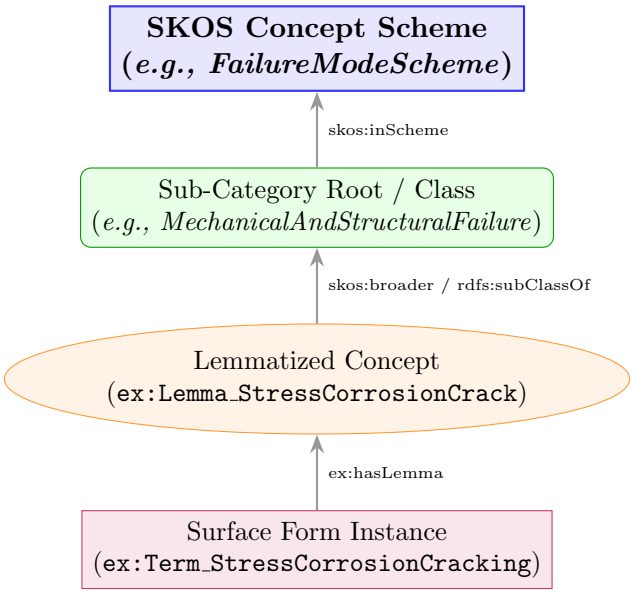


Figure 6: Four-tier hierarchical structure of entities within the Pipeline Risk Ontology.

Automated Sub-Categorization Engine To establish deeper taxonomic trees, terms are programmatically routed into sub-categories using regular expression pattern matching and keyword heuristics within the build script:

- **Failure Modes taxonomy:** Functional categorization includes sub-roots for *Corrosion & Electrochemical Failure*, *Mechanical & Structural Failure*, *Geotechnical & Ground Failure*, *Hydraulic & Operational Surge Failure*, and *Third-Party Interference*.
- **Risk Variables taxonomy:** Implements an 8-class classification hierarchy (`ex:VariableClassific`
  1. Meteorological & Hydrological Monitoring
  2. Hydrological Settings & Flood Risk
  3. Geotechnical Dynamics & Landslide Hazards
  4. Pedological Soil Properties & Agricultural Capacity
  5. Topographic & Climatic Indicators
  6. Built Environment, Land Use, & Human Interventions
  7. Landscape Regionalization
  8. Socio-Economic Indicators, Vulnerability, & Municipal Metabolism
- **Causal Relations taxonomy:** Structured into *Direct Causation*, *Conditional/Contributing*, *Correlation/Association*, and *Attribution/Derivation*.

Knowledge Graph Serialization The fully constructed graph links each surface instance to its conceptual lemma via `ex:hasLemma`, maps lemmas to broader category concepts via `skos:broader` and `rdfs:subClassOf`, and binds top-level concepts to their respective SKOS schemes via `skos:inScheme`. The resulting RDF graph is serialized into both standard W3C Turtle (`.ttl`) and RDF/XML (`.rdf`) formats for downstream inference and SPARQL query processing.

4 Methodology

4.1 Overview

Designing a data-driven system capable of anticipating possible water system failures requires bridging distinct domains, for this approach: observed physical reality and codified technical knowledge. While geospatial data captures the physical and environmental states of places where water infrastructure networks across fine spatial and temporal resolutions are present, patent literature archives decades of engineered failure mechanisms, material stresses, and documented causal relationships. This chapter presents a methodology for fusing linguistic patent knowledge with geospatial intelligence. By integrating these disparate data streams into a unified analytics pipeline, the system transcends reactive monitoring, establishing a predictive framework that links observable environmental triggers to documented technical failure modes. The architecture is engineered as a scalable, automated pipeline that ingests continuous geospatial updates and periodically refreshes patent knowledge bases, ensuring that risk assessment remains adaptive to both environmental change and evolving technical innovations.

4.2 Conceptual Pipeline

The conceptual pipeline rests on a dual-stream architecture that harmonizes spatial environmental variables with textual engineering records. The spatial stream ingests multi-source geospatial measurements, normalizes them onto a uniform discrete provincial grid, and performs spatial correlation analyses to identify localized activation zones where environmental stressors co-occur. Concurrently, the patent processing stream executes domain-specific queries, cleans textual corpora down to sentence-level technical statements, and extracts structured causal relationships. These extracted causal triplets—comprising technical terms, spatial variables, failure modes, and causal links—are vectorized via sentence transformers and mapped semantically to the spatial domain. The final integration layer correlates spatial activation zones with patent-derived causal chains, producing a validated, predictive risk assessment framework that continuously updates as new operational and textual data streams become available.

As depicted in Figure 7, the geospatial stream ingests multi-source environmental measurements, normalizes them onto a uniform discrete provincial grid, and performs spatial correlation analyses to identify localized activation zones where environmental stressors co-occur. Concurrently, the patent processing stream executes domain-specific queries, cleans textual corpora down to sentence-level technical statements, and extracts structured causal relationships. These extracted causal triplets—comprising technical terms, spatial variables, failure modes, and causal links—are vectorized via sentence transformers and mapped semantically. The final integration layer correlates spatial activation zones with patent-derived causal chains, producing a validated, predictive risk assessment framework that continuously updates as new operational and textual data streams become available.

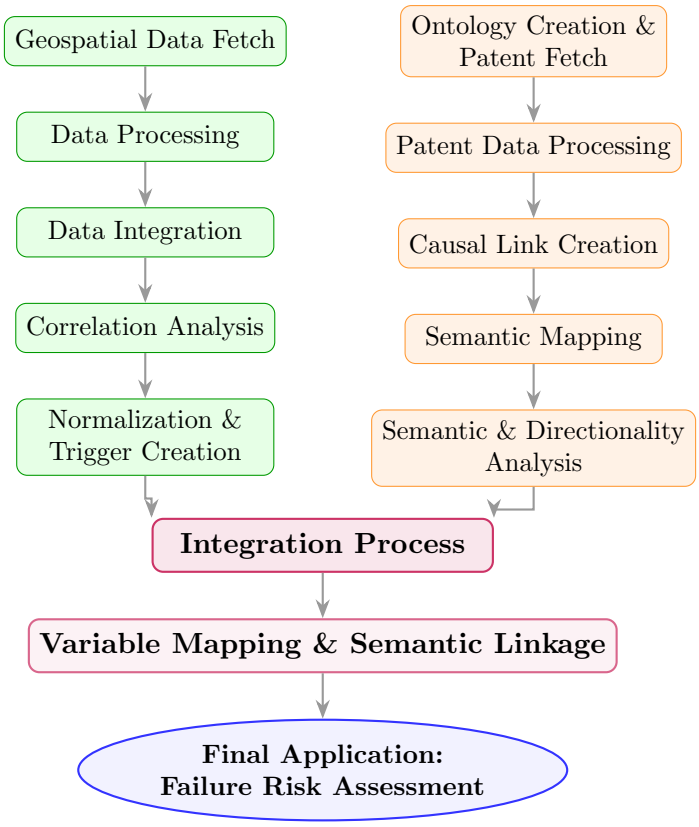


Figure 7: Downstream conceptual schema of the dual-stream geospatial and patent integration architecture.

4.3 Data Flow

The data flow within the system is structured as a directed acyclic graph that governs information from raw ingestion to final risk scoring. Ingestion processes handle spatial inputs operating at hourly, daily, or monthly frequencies alongside periodic monthly updates to the patent repository. Spatial ingestion feeds raw points, lines, and polygons through a sequential preprocessing pipeline consisting of spatial joins, line integrations, polygon overlays, and centroid derivations, culminating in a finalized geospatial data cube. Simultaneously, the text mining stream queries patent databases, strips extraneous boilerplate, and applies semantic parsing to isolate causal sentences. These streams converge during the spatial-patent linking phase, where semantic embeddings bridge textual descriptions with spatial variables. The integrated output is stored in downstream analytical tables, feeding predictive models that evaluate water system vulnerability in real time.

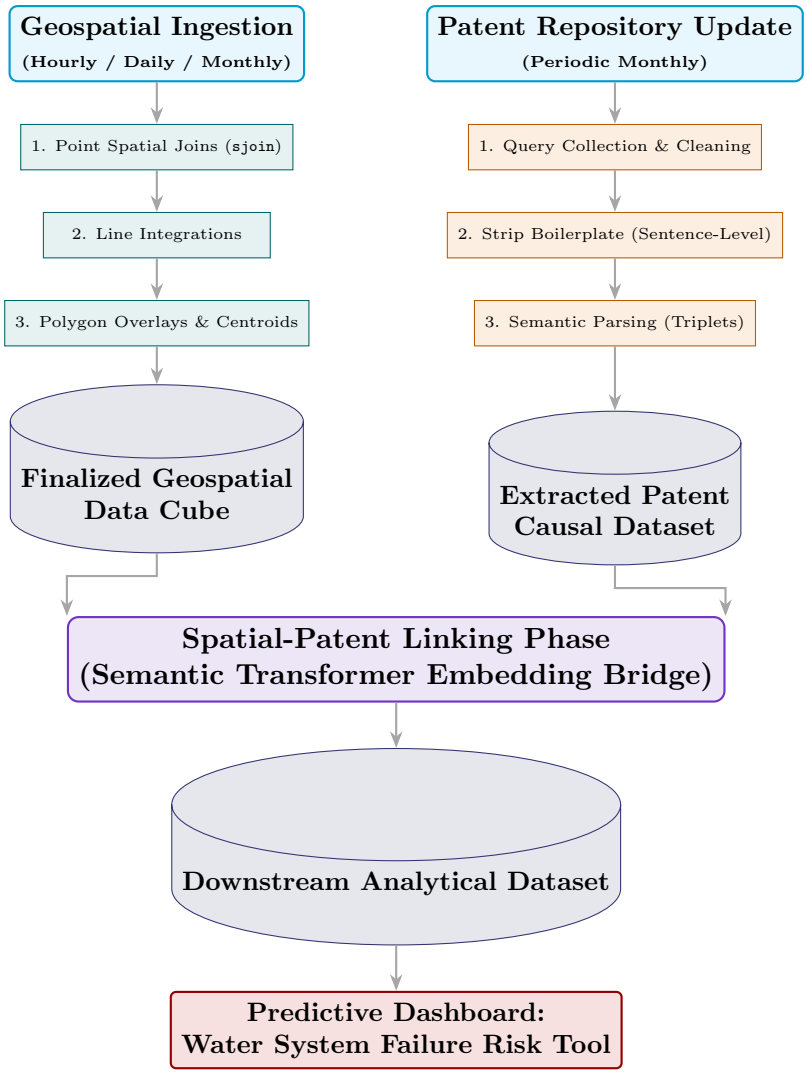


Figure 8: Directed Acyclic Graph (DAG) detailing the flow of spatial and textual data from ingestion to real-time failure prediction.

4.4 Design Considerations

Building a hybrid geospatial-patent risk framework entails addressing several structural, data, and computational challenges. Operating exclusively with free and open data sources imposes operational constraints, necessitating a lightweight, modular pipeline architecture capable of seamlessly adapting to new or modified datasets as they become available. Furthermore, given the absence of proprietary pipeline or detailed water distribution network data, achieving actionable spatial intelligence requires a fine resolution. The Uber H3 hexagonal hierarchical spatial index at resolution 10, providing approximately one-hectare cells, was selected to establish localized zones where field inspections and infrastructure repairs can be carried out in an efficient and scalable manner, avoiding the edge-distortion artifacts common in square grid systems. Another critical consideration is data distribution asymmetry: geospatial variables frequently exhibit heavy-tailed nominal values that can disproportionately skew correlation metrics, prompting the enforcement of logarithmic scaling combined with Yeo-Johnson transformations to stabilize variance. To maintain pipeline scalability for larger geographic extents and alternative risk domains, patent analysis was intentionally restricted to textual processing in this initial phase, bypassing complex graphical network extraction. Finally, while socioeconomic variables were introduced to test whether they meaningfully interacted with and impacted high-risk spatial areas, ultimate design priorization established that environmental factors are the most vital and relevant drivers governing the system.

4.5 Data Collection and Preparation

4.5.1 Geospatial Data Processing

The foundation of the physical risk assessment model relies on ingesting, cleaning, and standardizing diverse geospatial layers. Raw spatial features are harvested from multiple institutional and open-source monitoring databases. Because these datasets arrive in heterogeneous formats and coordinate systems, a rigorous spatial normalization protocol is executed to align all features onto an H3 hexagonal grid at dimension 10, yielding cell areas of approximately one hectare. This normalization ensures consistent spatial joins across disparate monitoring layers.

The spatial integration workflow proceeds through a sequential series of modular Python scripts. The process initiates with point-based feature integration via *1_integrate_points_sjoin.py*, where monitoring stations and localized incident points are spatially joined to the reference grid. Linear infrastructure, such as waterways, canals, and river networks, is incorporated through *2_integrate_lines.py*, mapping linear density and intersection events to respective H3 cells. Complex polygonal boundaries, including watershed catchments and land-use zones, are processed via *3_integrate_polygons_overlay.py* and *4_integrate_polygons_centroid.py* to capture areal coverage and central locality markers. Water station telemetry is merged via *5_integrate_stations.py*, bringing in temporal observations. Finally, *6_finalize_datacube.py* compiles these disparate streams into a unified, geospatial data cube.

4.5.2 Patent Data Processing

Complementing the geospatial data cube, the textual stream extracts engineering intelligence from global patent repositories. The patent processing pipeline begins with the construction of a pre-established domain ontology containing hierarchical term structures for water infrastructure, environmental stressors, and mechanical components. A targeted query-based patent collection phase retrieves relevant technical documents. The retrieved corpus undergoes rigorous

cleaning to discard administrative boilerplate, retaining only sentence-level content containing keywords of interest.

From this filtered corpus, the system constructs causal links by parsing sentences into standardized structural triplets. Each extracted triplet consists of a technical word, an attribute or spatial variable, a specific failure mode, and a causal connecting word bridging the variable to the failure defect. By identifying these causal chains within patent abstracts and claims, the system abstracts documented engineering failure mechanisms into machine-readable structures that mirror physical environmental stressors.

4.6 Spatial Data Analysis

4.6.1 Spatial Correlation Analysis

To determine which environmental variables act as compounding stressors, the spatial data cube undergoes correlation analysis across the H3 grid cells. Pairs of variables that frequently co-occur spatially are identified to uncover potential multi-factor triggers of water system failures. Because environmental and infrastructural variables often span vastly different numerical scales and exhibit severe skewness, robust statistical transformations are applied.

A log1p and Yeo-Johnson normalization is first applied to dampen the impact of extreme nominal values, and stabilize variance across asymmetric distributions. Following normalization, a strict co-occurrence threshold is applied to filter out background noise, retaining only the most statistically significant variable pairs. These highly correlated variables define localized activation zones within the spatial map, representing high-risk operational environments where system vulnerabilities are most likely to manifest.

4.7 Patent Data Analysis

4.7.1 Causal Relation Extraction

The system identifies potential causal chains by forming structured triplets that bind technical terms, spatial or material variables, and failure modes together via explicit causal connectors. To map these unstructured textual descriptions to standardized system variables, sentence transformers generate dense semantic embeddings for both terms and topics.

Furthermore, the pipeline conducts a fine-grained directionality analysis on the extracted causal statements. This analysis distinguishes between verb-mediated causes and non-verb associations, while evaluating the argumentative tone of the text to differentiate between informative, objective technical descriptions and passive or speculative claims. This ensures that only high-confidence, empirically grounded engineering relationships are propagated to the integration stage.

4.7.2 Ontology Integration

To anchor the extracted patent insights within a rigorous domain taxonomy, the pre-established ontology structures are enforced as semantic filters. Ontology terms govern the assignment of specific technical attributes to each extracted concept. By mapping patent entities directly to the ontology's hierarchical tree, the system constructs reliable patent links that connect generalized failure modes to specific infrastructure components.

4.8 Integration Process

4.8.1 Spatial-Patent Linking

The core methodological innovation of this research lies in bridging the spatial and textual domains. Every validated patent causal chain is linked to its corresponding physical variable within the spatial map. This connection is established via the semantic linkages derived during the embedding phase, aligning textual failure descriptions with their corresponding environmental triggers in the H3 data cube.

4.8.2 Final Integration

The culmination of the framework is an end-to-end automated processing pipeline that unifies spatial and textual streams. Spatial triggers derived from H3 activation zones are dynamically correlated with patent-derived causal chains, producing a comprehensive predictive model for water system failures. This architecture ensures that observed environmental anomalies are immediately contextualized by documented engineering failure knowledge.

4.9 Pipeline Architecture

4.9.1 Data Ingestion

The system architecture is planned for continuous, automated execution. The spatial ingestion module accommodates multi-frequency data feeds, accepting hourly weather station readings, daily operational updates, and monthly GIS layer refreshes. Concurrently, the patent data ingestion module executes scheduled monthly queries to fetch newly published patents, ensuring that the system's technical knowledge base remains current.

4.9.2 Processing Workflow

The processing workflow operates as a continuous Extract, Transform, Load pipeline governed by automated scheduling. When new spatial data arrives, the H3 grid integration, correlation analysis, and activation zone generation routines execute automatically. Similarly, when the patent database refreshes, the text cleaning, causal chain extraction, and ontology mapping routines re-run. Finally, the integration engine updates the spatial-patent semantic links, recalculating risk scores and refreshing predictive failure outputs.

4.10 Output and Applications

4.10.1 Spatial Triggers

The spatial outputs consist of empirically identified failure indicators and localized activation zones where high correlations of stressor variables converge, marking high-risk operational environments within the water network.

4.10.2 Patent-Driven Insights

The patent subsystem yields structured causal chains and documented failure modes, offering deep technical insight into the physical and mechanical conditions that precipitate infrastructure breakdown.

4.10.3 Integrated Results

By synthesizing spatial triggers with patent evidence, the integrated system delivers predictive failure capabilities. This risk assessment framework provides water authorities with a data-driven tool to anticipate system failures before physical manifestations occur, bridging empirical environmental monitoring engineering knowledge. FILL THE SUMMARY STATISTICS FOR THE DATA, THE PREVIOUS CHAPTERS ARE ALREADY VERIFIED

5 Implementation and Experimental Setup

5.1 Implementation Overview

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.2 Implementation Details

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.3 Experimental Setup

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.4 Evaluation Protocol

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.5 Experimental Scenarios

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci

eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.6 Evaluation Metrics

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6 Results and Analysis

6.1 Overview

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.2 Empirical Findings

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.2.1 Data Characteristics

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.2.2 Performance Metrics

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit

amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.2.3 Comparison with State-of-the-Art

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.2.4 Statistical Significance

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.3 Business & Technical Interpretation

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.3.1 Technical Interpretation

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.3.2 Business Interpretation

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit

amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.4 Model Explainability & Limitations

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.4.1 Model Explainability

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.4.2 Limitations

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.5 Scenario Analysis: A Business Case Study

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

6.6 Summary of Key Results

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci

eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

7 Conclusion and Future Work

7.1 Summary of Contributions

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

7.2 Final Business and Technical Takeaways

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

7.3 Recommendations for Future Work

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

References

[1] A. Mithal and B. Thankachan, "A comprehensive review on blockchain-based systems for groundwater conservation and wastewater management," *Environmental Management*, vol. 75, 2025. [Online]. Available: https://doi.org/10.1007/s00267-025-02247-6

[2] F. Utilitatis and Utilitalia, "Blue book 2026: I dati del servizio idrico integrato in italia," Fondazione Utilitatis, Roma, Italia, Rapporto Annuale, 2026.

[3] I. Figueiredo, P. Esteves, and P. Cabrita, "Water wise – a digital water solution for smart cities and water management entities," *Procedia Computer Science*, vol. 181, 2021. [Online]. Available: https://doi.org/10.1016/j.procs.2021.01.245

[4] E. A. Saleh, T. Zayed, W. Xu, L. Xie, and A. S. Mohammed, "Scalable detection of underground water leaks in urban networks using l-band sar imagery and machine learning," Hong Kong Supplies Department, Tech. Rep., 2024.

[5] T. P. Raptis, A. Passarella, and M. Conti, "Data management in industry 4.0: State of the art and open challenges," *IEEE Access*, vol. 7, 2019. [Online]. Available: https://doi.org/10.1109/ACCESS.2019.2929296

[6] G. Antzoulatos, C. Mourtzios, P. Stournara *et al.*, "Making urban water smart: the smart-water solution," *Water Science & Technology*, vol. 82, no. 12, 2020. [Online]. Available: https://doi.org/10.2166/wst.2020.391

[7] M. Singh and S. Ahmed, "Iot based smart water management systems: A systematic review," *Materials Today: Proceedings*, vol. 46, 2021. [Online]. Available: https://doi.org/10.1016/j.matpr.2020.08.588

[8] S. Baimukhanov, H. Ali, and A. Yazici, "Enhancing ml-based anomaly detection in data management for security on integration of iot, cloud computing, and edge computing," *Expert Systems With Applications*, vol. 293, 2025. [Online]. Available: https://doi.org/10.1016/j.eswa.2025.128700

[9] C. O. Klingenberg *et al.*, "Industry 4.0 as a data-driven paradigm: a systematic literature review," *Journal of Manufacturing Technology Management*, vol. 32, no. 3, 2021. [Online]. Available: https://doi.org/10.1108/JMTM-09-2018-0325

[10] J. Lioumbas, T. Spahos, A. Christodoulou, I. Mitzias, P. Stournara, I. Kavouras, A. Mentes, N. Theodoridou, and A. Papadopoulos, "Multi-component remote sensing for mapping buried water pipelines," *Remote Sensing*, vol. 17, 2025. [Online]. Available: https://doi.org/10.3390/rs17122109

[11] S. Kamm, N. Jazdi, and M. Weyrich, "Knowledge discovery in heterogeneous and unstructured data of industry 4.0 systems: Challenges and approaches," *Procedia CIRP*, vol. 104, 2021. [Online]. Available: https://doi.org/10.1016/j.procir.2021.11.164

[12] M. S. Rahim, K. A. Nguyen, R. A. Stewart, D. Giurco, and M. Blumenstein, "Machine learning and data analytic techniques in digital water metering: A review," *Water*, vol. 12, 2020. [Online]. Available: https://doi.org/10.3390/w12010294

[13] R. U. Hincapie, "Enhancing water transport efficiency through geospatial risk modeling," Master's Thesis Presentation Slides, University of Pisa, 2026.

[14] U. Sangroula, V. Viñas, M. Odhiambo, and T. J. R. Pettersson, "Integrating machine learning and multi-criteria decision analysis for health risk management in water distribution networks," *Scientific Reports*, vol. 16, 2026. [Online]. Available: https://doi.org/10.1038/s41598-026-52465-z

[15] N. Donratanapat, S. Samadi, J. M. Vidal, and S. Sadeghi Tabas, "A national scale big data analytics pipeline to assess the potential impacts of flooding on critical infrastructures and communities," *Environmental Modelling & Software*, vol. 133, 2020. [Online]. Available: https://doi.org/10.1016/j.envsoft.2020.104828

[16] L. Zhang, Y. Xie, L. Xidao, and X. Zhang, "Multi-source heterogeneous data fusion," National University of Defense Technology, Tech. Rep., 2025.

[17] J. Zhu, Y. Mao, L. Chen, C. Ge, Z. Wei, and Y. Gao, "Fusionquery: On-demand fusion queries over multi-source heterogeneous data," *Proceedings of the VLDB Endowment*, vol. 17, no. 6, 2024. [Online]. Available: https://doi.org/10.14778/3648160.3648174

[18] B. Yu, C. Zhang, Y. Li, and J. Sun, "A general multi-source data fusion framework based on rdf," in *Proceedings of ICMLC*, 2019. [Online]. Available: https://doi.org/10.1145/3318299.3318394

[19] J. Yan, Y. Meng, L. Lu, and L. Li, "Industrial big data in an industry 4.0 environment: Challenges, schemes, and applications for predictive maintenance," *IEEE Access*, vol. 5, 2017. [Online]. Available: https://doi.org/10.1109/ACCESS.2017.2765544

[20] S. R. Newrzella, D. W. Franklin, and S. Haider, "5-dimension cross-industry digital twin applications, model, and analysis," *IEEE Access*, vol. 9, 2021. [Online]. Available: https://doi.org/10.1109/ACCESS.2021.3115055

[21] Q. Qi, F. Tao, T. Hu, N. Anwer, A. Liu, Y. Wei, L. Wang, and A. Y. C. Nee, "Enabling technologies and tools for digital twin," *Journal of Manufacturing Systems*, vol. 58, 2021. [Online]. Available: https://doi.org/10.1016/j.jmsy.2019.10.001

[22] B. Li, D. Pi, Y. Lin, I. A. Khan, and L. Cui, "Multi-source information fusion based heterogeneous network embedding," *Information Sciences*, 2020. [Online]. Available: https://doi.org/10.1016/j.ins.2020.05.012

[23] B. Li *et al.*, "Multi-source information fusion for spatio-temporal graph cnns," *Information Sciences*, 2020.

[24] J. Ngiam *et al.*, "Multimodal deep learning," in *Proceedings of the 28th International Conference on Machine Learning (ICML)*, 2011.

[25] V. T. Enterprise GIS Team, "Collection and compilation of water pipeline field performance," Virginia Tech Water Resources Division, Tech. Rep., 2024.

[26] A. Malekloo, E. Ozer, M. AlHamaydeh, and M. Girolami, "Machine learning and structural health monitoring overview with emerging technology and high-dimensional data source highlights," *Structural Health Monitoring*, vol. 21, no. 4, 2021. [Online]. Available: https://doi.org/10.1177/1475921721101608

[27] V. Giordano, F. Chiarello, N. Melluso, G. Fantoni, and A. Bonaccorsi, "Text and dynamic network analysis for measuring technological convergence: A case study on defense patent data," *IEEE Transactions on Engineering Management*, 2021. [Online]. Available: https://doi.org/10.1109/TEM.2021.3078231

[28] J. Jee, S. Park, and S. Lee, "Patent analysis using patent image data: Technology intelligence, product structure, and network analysis," *Journal of Informetrics*, vol. 16, 2022. [Online]. Available: https://doi.org/10.1016/j.joi.2022.101263

[29] K. Pustu-Iren, G. Bruns, and R. Ewerth, "A multimodal approach for semantic patent image retrieval," in *CEUR Workshop Proceedings*, vol. 2909, 2021.

[30] L. Li *et al.*, "Extraction, recognition, and fusion of design patent image features," *Multimedia Tools and Applications*, vol. 83, 2024.

[31] M. Singh *et al.*, "Flood and landslide prediction and risk assessment system using spatial-sequential networks," *International Journal of Disaster Risk Science*, vol. 13, 2024.

[32] M. U. S. Khan, A. Y. Battamo, R. Ravindar, and M. Salauddin, "The role of artificial intelligence in bathing water quality assessment: Trends, challenges, and opportunities," *Water*, vol. 17, no. 21, 2025. [Online]. Available: https://doi.org/10.3390/w17213176

[33] J. S. Lioumbas, A. Christodoulou, and A. Mentes, "Exploring multitemporal remote sensing for subterranean feature detection," in *Proceedings of SPIE*, vol. 13816, 2024.

[34] Y. Qu *et al.*, "Probabilistic assessment of water network influence zones," *Big Earth Data*, vol. 10, no. 1, 2026. [Online]. Available: https://doi.org/10.1080/20964471.2025.2597579

[35] S. Radini *et al.*, "Risk management and digitalisation for safe water reuse in agriculture," *Journal of Water Process Engineering*, vol. 53, 2023. [Online]. Available: https://doi.org/10.1016/j.jwpe.2023.103690

[36] X. Li *et al.*, "A graph neural network-based stock forecasting method utilizing multi-source heterogeneous data fusion," *PLOS ONE:Multimedia Tools and Applications*, vol. 81, 2022.

[37] J. Wang, H. Jia, and T. Peng, "A graph-based approach to multi-source heterogeneous information fusion in stock market," *PLoS ONE*, vol. 17, no. 8, 2022. [Online]. Available: https://doi.org/10.1371/journal.pone.0272083

[38] R. X. Gao, J. Krüger, M. Merklein, H.-C. Möhring, and J. Vancza, "Artificial intelligence in manufacturing: State of the art and future directions," *CIRP Annals - Manufacturing Technology*, vol. 73, no. 2, 2024. [Online]. Available: https://doi.org/10.1016/j.cirp.2024.04.101

[39] N. V. Golovastikov, N. L. Kazanskiy, and S. N. Khonina, "Optical fiber-based structural health monitoring: Advancements, applications, and integration with artificial intelligence for civil and urban infrastructure," *Photonics*, vol. 12, 2025. [Online]. Available: https://doi.org/10.3390/photonics12060615

[40] H. G. P. Gowda, H. N. Nisarga, P. P. Mattihalli, G. K. Bhoomika, and A. Geetha Kiran, "Hydrosense 2.0: An iot-based smart water quality monitoring and alert system with machine learning and cloud integration," *International Journal of Innovative Research in Technology*, vol. 11, 2025.

[41] T. Shen *et al.*, "A study on automatic annotation methods for watershed," *European Journal of Remote Sensing*, vol. 58, no. 1, 2025. [Online]. Available: https://doi.org/10.1080/22797254.2025.2473939

[42] L. Duan and L. D. Xu, "Data analytics in industry 4.0: A survey," *Information Systems Frontiers*, vol. 26, 2024. [Online]. Available: https://doi.org/10.1007/s10796-021-10190-0

[43] S. Ghosh *et al.*, "Leveraging directed acyclic graphs for advanced ensemble learning in predictive maintenance for industry 4.0," in *Proceedings of the IEEE*, 2025.

[44] Y. Xie *et al.*, "Ai-enabled smart wastewater treatment plants: A comprehensive review," *Journal of Environmental Management*, vol. 401, 2026. [Online]. Available: https://doi.org/10.1016/j.jenvman.2026.128949

[45] dati.gov.it, "Linee guida per i cataloghi dati nazionali: Ricerca termini acqua, rete idrica, e servizio idrico integrato," National Open Data Portal, 2025. [Online]. Available: https://www.dati.gov.it

[46] A. di Regolazione per Energia Reti e Ambiente (ARERA), "Dati e statistiche sul servizio idrico integrato," ARERA Official Portal, 2025. [Online]. Available: https://www.arera.it/dati-e-statistiche?ambito=36

[47] S. N. per la Protezione dell'Ambiente (SNPA), "I nodi del sistema - elenco dei siti web delle arpa e appa," SNPA Portal, 2025. [Online]. Available: https://www.snpambiente.it/chi-siamo/i-nodi-del-sistema/i-siti-web

[48] A. I. T. (AIT), "Governance del servizio idrico integrato: Comuni, conferenze territoriali e gestori in toscana," Regione Toscana Open Data, 2025. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=8288a32b-3756-40d3-a41d-959575c44e57

[49] Eurostat, "Water statistics at the national level (env_wat_res)," European Commission, 2024. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/env_wat_res/default/table

[50] Eurostat, "Renewable water resources (national) (env_wat_res)," European Commission, 2024. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/env_wat_res/default/table

[51] Eurostat, "Sewage sludge production and disposal in europe (env_ww_spd)," European Commission, 2024. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/env_ww_spd/default/table

[52] E. GISCO, "Geospatial reference data grids: Elevation, transportation, and land use," European Commission, 2024. [Online]. Available: https://ec.europa.eu/eurostat/web/gisco/geodata/grids

[53] Copernicus, "Copernicus data space ecosystem earth observation portal," European Space Agency, 2025. [Online]. Available: https://dataspace.copernicus.eu

[54] C. C. C. S. (C3S), "Era5 hourly atmospheric and climate reanalysis variables," European Centre for Medium-Range Weather Forecasts (ECMWF), 2025. [Online]. Available: https://climate.copernicus.eu/climate-reanalysis

[55] C. L. M. S. (CLMS), "Sentinel-2 crop classification and land cover product suite," European Commission, 2024. [Online]. Available: https://land.copernicus.eu/en/map-viewer

[56] J. R. C. J. ESDAC, "European soil database soil properties, functions and point data," European Soil Data Centre (ESDAC), 2024. [Online]. Available: https://esdac.jrc.ec.europa.eu/resource-type/european-soil-database-soil-properties

[57] I. N. di Statistica (ISTAT), "Esploradati istat portal - environmental and water indicators," ISTAT Data Portal, 2025. [Online]. Available: https://esploradati.istat.it/databrowser/

[58] I. N. di Statistica (ISTAT), "Water withdrawal for public supply (prelacq 1)," ISTAT Databrowser, 2024. [Online]. Available: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0920ENV,1.0/ENV_WATER/IT1,12_340_DF_DCCV_PRELACQ_1,1.0

[59] I. N. di Statistica (ISTAT), "Public water supply use in italian municipalities (consacqua 2)," ISTAT Databrowser, 2024. [Online]. Available: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0920ENV,1.0/ENV_WATER/DCCV_CONSACQUA/IT1,12_60_DF_DCCV_CONSACQUA_2,1.0

[60] I. N. di Statistica (ISTAT), "Geological data: Municipalities and areas at hydrogeological, landslide, and flood risk," ISTAT Databrowser, 2024. [Online]. Available: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0930TER,1.0/DCCV_CARGEOMOR_COM/IT1,DCCV_CARGEOMOR_COM,1.0

[61] A. di Regolazione per Energia Reti e Ambiente (ARERA), "Componenti della spesa media per il servizio idrico integrato (sii)," ARERA Portal, 2024. [Online]. Available: https://www.arera.it/dati-e-statistiche/dettaglio/componenti-della-spesa-media

[62] A. di Regolazione per Energia Reti e Ambiente (ARERA), "Automatic compensation and quality remedies by user type (indennizzi automatici)," ARERA Portal, 2024. [Online]. Available: https://www.arera.it/dati-e-statistiche/dettaglio/indennizzi-automatici-per-tipologia-di-utenza

[63] A. di Regolazione per Energia Reti e Ambiente (ARERA), "Integrated water service contractual quality regulation (rqsii) performance data," ARERA Portal, 2021. [Online]. Available: https://www.arera.it/dati-e-statistiche/dettaglio/rqsii

[64] A. . ISPRA, "Sewerage system adequacy m4 performance indicator gml dataset," RNDT National Catalog, 2024. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=962c5851-c072-445a-9e55-325a39d3ce4f

[65] A. . ISPRA, "Water losses in national aqueducts m1 performance indicator gml dataset," RNDT National Catalog, 2024. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=cf8aa6f0-fcc5-49e4-9207-d27f40d4fa1c

[66] R. N. dei Dati Territoriali (RNDT), "Polygonal layer of the public water domain (demanio idrico) boundaries," RNDT National Portal, 2025. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=ad767b67-95cc-441e-8fc1-b24efd9097ff

[67] R. N. dei Dati Territoriali (RNDT), "Strahler water element stream order classification (dbgt 10k)," RNDT Catalog, 2025. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=16c4b5e1-0386-4551-a56e-ae7927d59ec2

[68] R. N. dei Dati Territoriali (RNDT), "Dbtr - hydrological connection node - (noi_gpt)," RNDT Portal, 2025. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=a5bf827a-a95e-4edf-8dff-934533c166bb

[69] R. N. dei Dati Territoriali (RNDT), "Dbtr2008 - historical hydrological connection node - (noi_gpt)," RNDT Portal, 2008. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=4624ef40-43bf-42ee-9a1b-1ce0e22d22f1

[70] C. Association, "Ckan api specification v3: Package search and metadata harvesting rest endpoints," CKAN Documentation, 2025. [Online]. Available: https://docs.ckan.org/en/latest/api/

[71] "Mass fetching metadata synthesis: Regional, format, cadence, and domain distribution of italian open water data," Italian Water and Energy Infrastructure Data Directory, Automated API Harvesting Technical Report, 2025.

[72] dati.gov.it, "Repertorio nazionale dei dati aperti: Ckan api endpoint e catalogo nazionale," AgID - Agenzia per l'Italia Digitale, 2025. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset

[73] S. S. Organizations, "Statistical data and metadata exchange (sdmx) restful api specification," SDMX Official Portal, 2024. [Online]. Available: https://sdmx.org

[74] I. N. di Statistica (ISTAT), "Confini delle unità amministrative a fini statistici al 1° gennaio 2018," 2018, accessed: 2026-09-20. [Online]. Available: https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

[75] I. N. di Statistica (ISTAT), "Basi territoriali e variabili censuarie," 2021, accessed: 2026-09-20. [Online]. Available: https://www.istat.it/notizia/basi-territoriali-e-variabili-censuarie/

[76] I. S. per la Protezione e la Ricerca Ambientale (ISPRA), "Piattaforma idrogeo - open data dissesto idrogeologico," 2024, accessed: 2026-09-20. [Online]. Available: https://beta.idrogeo.isprambiente.it/app/page/open-data

[77] I. S. per la Protezione e la Ricerca Ambientale (ISPRA), "Rendis - repertorio nazionale degli interventi per la difesa del suolo," 2024, accessed: 2026-09-20. [Online]. Available: https://www.rendis.isprambiente.it/rendisweb/vistepub.jsp

[78] R. Toscana, "Geoscopio - cartoteca digitale della regione toscana," 2024, accessed: 2026-09-20. [Online]. Available: https://www502.regione.toscana.it/geoscopio/cartoteca.html#

[79] R. Toscana, "Catalogo open data regione toscana," 2024, accessed: 2026-09-20. [Online]. Available: https://dati.toscana.it/dataset

[80] I. N. di Geofisica e Vulcanologia (INGV), "Cpti15 - catalogo parametrico dei terremoti italiani e dbmi15 - database macroseismico italiano," 2021, accessed: 2026-09-20. [Online]. Available: https://emidius.mi.ingv.it/CPTI15-DBMI15/

[81] S. I. e Geotecnico Regionale (SIR Toscana), "Consistenza della rete meteo-idrografica regionale," 2024, accessed: 2026-09-20. [Online]. Available: https://www.sir.toscana.it/consistenza-rete

[82] R. Toscana, "Dbped - banca dati pedologica della regione toscana," 2023, accessed: 2026-09-20. [Online]. Available: https://dati.toscana.it/dataset/dbped

[83] I. N. di Statistica (ISTAT), "Contribuenti e reddito complessivo per classi di importo (fonte mef)," 2023, accessed: 2026-09-20. [Online]. Available: https://esploradati.istat.it/databrowser/#/it/dw/categories/IT1,HOU,1.0/MEF_REDDITIIRPEF_COM/IT1,30_1008_DF_MEF_REDDITIIRPEF_COM_2,1.0

[84] I. N. di Statistica (ISTAT), "Indice composito di fragilità e sue componenti a livello comunale," 2023, accessed: 2026-09-20. [Online]. Available: https://esploradati.istat.it/databrowser/#/it/dw/categories/IT1,Z0930TER,1.0/CFI_MUN/IT1,DF_COMP_FRA_IND_MUNICIPAL_01,1.0

[85] I. N. di Statistica (ISTAT), "Statistiche sull'acquedotto e consumo acqua," 2023, accessed: 2026-09-20. [Online]. Available: https://esploradati.istat.it/databrowser/#/it/dw/categories/IT1,Z0920ENV,1.0/ENV_WATER/DCCV_CONSACQUA/IT1,12_60_DF_DCCV_CONSACQUA_2,1.0

[86] I. N. di Statistica (ISTAT), "Popolazione residente - serie storica censuaria per comune," 2023, accessed: 2026-09-20. [Online]. Available: https://esploradati.istat.it/databrowser/#/it/censpop/categories/IT1,Z1200CPA,1.0/CPA_POP/IT1,DF_DCSS_POPRES_SERIES_TV_1,1.0

[87] D. G. O. Data, "Sistema delle acque - specchi d'acqua," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=e94950e0-8fa0-4002-af34-8c4772579c16

[88] D. G. O. Data, "Sistema delle acque - corso d'acqua," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=845c7f23-caf9-4c39-be00-accd70b837d1

[89] D. G. O. Data, "Produzioni impianti anconella e mantignano," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=89297912-76bb-4670-bdcb-008238d93d64

[90] D. G. O. Data, "Rete dei fontanelli pubblici," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=5688216f-ed79-477c-b833-02b89c124eee

[91] D. G. O. Data, "Litri erogati fontanelli," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=dfc7649e-a2bf-4a37-ae59-b2eea21bbf05

[92] D. G. O. Data, "Livello invaso di bilancino," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=ff9be82a-a2c6-423a-8aea-88e82a82866f

[93] D. G. O. Data, "Tombinati - corso d'acqua sub-superficiali," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=1dfaec3e-5003-43f0-a80e-5395acfe5555

[94] D. G. O. Data, "Fontanelli alta qualità - attivi," 2023, accessed: 2026-09-20. [Online]. Available: https://www.dati.gov.it/view-dataset/dataset?id=f63e2c8e-97d0-4e59-ae71-9bda78464ab2

[95] O. Contributors, "Openstreetmap vector infrastructure data for florence province," 2024, extracted via QuickOSM plugin and Regional Technical Maps (CTR). [Online]. Available: https://www.openstreetmap.org/

[96] I. N. di Statistica (ISTAT), "Matrice del pendolarismo 2021," 2021, accessed: 2026-09-20. [Online]. Available: https://esploradati.istat.it/databrowser/#/it/censpop/categories/IT1,Z1200CPA,1.0/MATRICE_PEND

[97] A. di Bacino Distrettuale dell'Appennino Settentrionale, "Interferometria radar satellite sentinel-1 per il controllo dei movimenti del suolo," 2024, accessed: 2026-09-20. [Online]. Available: https://www.appenninosettentrionale.it/itc/?page_id=11499

[98] A. di Bacino Distrettuale dell'Appennino Settentrionale, "Arcgis webapp viewer per la deformazione del suolo insar," 2024, accessed: 2026-09-20. [Online]. Available: https://geodataserver.appenninosettentrionale.it/portal/apps/webappviewer/index.html?id=22257af5a31d40baae322a0e50b24151

[99] A. di Bacino Distrettuale dell'Appennino Settentrionale, "Arcgis rest portal service item - dati di dislocazione sentinel-1," 2024, accessed: 2026-09-20. [Online]. Available: https://geodataserver.appenninosettentrionale.it/portal/home/item.html?id=dfd20325529146dda50df608d4510045

[100] D. W. Cash, W. N. Adger, F. Berkes, P. G. Garden, L. Lebel, P. Olsson, L. Pritchard, and O. Young, "Scale and cross-scale dynamics: Governance and information in a multilevel world," *Ecology and Society*, vol. 11, no. 2, p. 8, 2006. [Online]. Available: https://ecologyandsociety.org

A Appendices

A.1 Supplementary Data

A.2 Additional Results

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

A.3 Technical Specifications

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Table 15: Summary Statistics (Numerical) (Sample Count: 225,000, Missing Values: 0)

Variable	Mean	Std	Variance	Min	5%	25%	50%	75%	95%	Max	Skewness
total_intervention_cost_1ring	5.8e+01	6.0e+03	3.6e+07	0	0	0	0	0	0	8.3e+05	1.2e+02
active_fountains_count_1ring	7.8e-04	2.8e-02	7.8e-04	0	0	0	0	0	0	1.0e+00	3.6e+01
total_fountains_count_1ring	7.6e-03	1.2e-01	1.4e-02	0	0	0	0	0	0	6.0e+00	2.0e+01
landslide_point_count_1ring	5.5e-01	9.1e-01	8.3e-01	0	0	0	0	1.0e+00	2.0e+00	1.1e+01	2.2e+00
tipo_movimento_<lambda_0>_1ring	2.0e+00	3.8e+00	1.5e+01	0	0	0	0	2.0e+00	1.1e+01	1.1e+01	1.8e+00
max_peak_elevation_1ring	1.4e+01	1.1e+02	1.1e+04	0	0	0	0	0	0	1.7e+03	8.4e+00
is_locality_1ring	1.2e-01	3.3e-01	1.1e-01	0	0	0	0	0	1.0e+00	1.0e+00	2.3e+00
seismic_event_count_1ring	2.1e-03	7.6e-02	5.8e-03	0	0	0	0	0	0	1.1e+01	1.0e+02
max_seismic_magnitude_1ring	7.9e-03	1.9e-01	3.7e-02	0	0	0	0	0	0	6.4e+00	2.5e+01
avg_seismic_magnitude_1ring	7.9e-03	1.9e-01	3.7e-02	0	0	0	0	0	0	6.4e+00	2.5e+01
road_density_m_per_m2	6.2e-03	9.4e-03	8.8e-05	0	0	0	2.1e-03	9.3e-03	2.2e-02	1.5e-01	3.2e+00
m_per_hex	9.7e+01	1.5e+02	2.1e+04	0	0	0	3.3e+01	1.5e+02	3.4e+02	2.4e+03	3.2e+00
dist_to_waterway_m	1.4e+02	1.3e+02	1.8e+04	7.0e-04	8.8e+00	4.6e+01	1.0e+02	1.9e+02	3.8e+02	1.5e+03	2.5e+00
rooting_depth_class	2.1e+00	1.2e+00	1.5e+00	0	1.0e+00	1.0e+00	2.0e+00	3.0e+00	4.0e+00	7.0e+00	5.2e-01
surface_stoniness_class	1.8e+00	1.0e+00	1.1e+00	0	1.0e+00	1.0e+00	2.0e+00	3.0e+00	4.0e+00	4.0e+00	5.3e-01
landslide_surface_class	2.4e+00	1.1e+00	1.2e+00	0	0	2.0e+00	3.0e+00	3.0e+00	4.0e+00	4.0e+00	-3.4e-01
avg_descending_soil_speed	-2.5e-02	6.1e-01	3.7e-01	-4.8e+01	0	0	0	0	0	7.0e+01	5.0e+00
avg_ascending_soil_speed	-1.9e-02	5.0e-01	2.5e-01	-2.9e+01	0	0	0	0	0	2.2e+01	-1.2e+01
hydraulic_hazard_level	3.9e-01	9.0e-01	8.0e-01	0	0	0	0	0	3.0e+00	3.0e+00	2.2e+00
epr_NTAXP	1.8e+04	4.6e+04	2.1e+09	8.7e+02	9.1e+02	3.7e+03	9.8e+03	1.4e+04	3.7e+04	2.8e+05	5.3e+00
epr_TAXABINC	1.7e+04	4.4e+04	2.0e+09	8.4e+02	8.9e+02	3.5e+03	9.5e+03	1.3e+04	3.6e+04	2.7e+05	5.3e+00
epr_CADINCR	1.7e+07	6.0e+07	3.6e+15	4.9e+05	5.1e+05	2.3e+06	6.3e+06	9.6e+06	2.9e+07	3.6e+08	5.5e+00
epr_CADINCF	8.3e+03	2.1e+04	4.5e+08	3.8e+02	4.4e+02	1.8e+03	4.4e+03	6.3e+03	1.8e+04	1.3e+05	5.3e+00
epr_SUBEMPTR	9.7e+03	2.5e+04	6.0e+08	4.1e+02	4.2e+02	1.9e+03	5.2e+03	7.6e+03	2.0e+04	1.5e+05	5.3e+00
epr_PENSINCR	1.3e+08	3.5e+08	1.2e+17	5.2e+06	6.7e+06	2.4e+07	5.6e+07	8.6e+07	2.7e+08	2.1e+09	5.4e+00
epr_PENSINCF	6.4e+03	1.6e+04	2.5e+08	3.4e+02	4.3e+02	1.5e+03	3.3e+03	4.8e+03	1.4e+04	9.6e+04	5.3e+00
epr_ENTROAIN	4.4e+01	8.4e+01	7.0e+03	0	0	1.4e+01	3.1e+01	4.2e+01	1.2e+02	5.0e+02	4.9e+00
epr_ENTROAIN01	2.3e+06	4.9e+06	2.4e+13	0	0	5.3e+05	1.1e+06	2.0e+06	4.9e+06	2.9e+07	4.9e+00
erd_E0_10000	4.2e+03	1.1e+04	1.1e+08	2.3e+02	2.6e+02	9.5e+02	2.2e+03	3.0e+03	8.5e+03	6.5e+04	5.3e+00
erd_E10000_1	2.3e+03	5.3e+03	2.8e+07	1.4e+02	1.4e+02	5.6e+02	1.2e+03	1.8e+03	4.9e+03	3.2e+04	5.3e+00
erd_E15000_2	5.8e+03	1.3e+04	1.7e+08	3.0e+02	3.4e+02	1.4e+03	3.4e+03	5.0e+03	1.3e+04	7.9e+04	5.2e+00
erd_E26000_5	4.4e+03	1.2e+04	1.5e+08	1.5e+02	1.6e+02	6.8e+02	2.1e+03	3.1e+03	9.5e+03	7.4e+04	5.4e+00
erd_E55000_7	5.3e+02	1.8e+03	3.2e+06	1.0e+01	1.7e+01	5.6e+01	1.7e+02	3.1e+02	8.7e+02	1.1e+04	5.5e+00
erd_E75000_1	4.0e+02	1.4e+03	2.1e+06	6.0e+00	1.0e+01	4.4e+01	1.2e+02	2.1e+02	5.9e+02	8.6e+03	5.5e+00
erd_E_GE1200	2.0e+02	7.5e+02	5.6e+05	0	0	1.3e+01	6.7e+01	8.8e+01	3.3e+02	4.5e+03	5.5e+00
edst_acq_imm	2.7e+03	8.3e+03	6.9e+07	0	8.9e+01	4.3e+02	1.3e+03	1.8e+03	5.2e+03	5.0e+04	5.4e+00
edst_acq_erog	1.5e+03	4.6e+03	2.1e+07	0	6.4e+01	3.1e+02	6.9e+02	1.0e+03	3.0e+03	2.8e+04	5.4e+00
eidx_COMP_FRA	3.3e+00	1.5e+00	2.2e+00	1.0e+00	1.0e+00	2.0e+00	3.0e+00	4.0e+00	6.0e+00	6.0e+00	3.5e-01
eidx_LAND_CON	7.3e+00	7.5e+00	5.7e+01	1.5e+00	1.5e+00	3.2e+00	5.2e+00	8.7e+00	2.0e+01	4.2e+01	3.2e+00
eidx_EMPL_RAT	7.3e+01	1.6e+00	2.5e+00	6.9e+01	7.0e+01	7.2e+01	7.3e+01	7.4e+01	7.6e+01	7.7e+01	6.5e-02
eidx_POP_25_6	3.8e+01	4.9e+00	2.4e+01	2.5e+01	2.6e+01	3.5e+01	3.9e+01	4.2e+01	4.4e+01	4.6e+01	-9.3e-01
eidx_POP_DEPE	7.5e+01	5.7e+00	3.2e+01	6.1e+01	6.8e+01	7.1e+01	7.3e+01	7.8e+01	8.6e+01	8.6e+01	4.7e-01
eidx_INDEX_AC	3.0e+01	1.5e+01	2.2e+02	0	0	2.1e+01	3.0e+01	4.2e+01	5.4e+01	5.4e+01	-3.0e-01
eidx_PERSEMP	7.6e+00	3.2e+00	1.0e+01	2.0e+00	3.0e+00	5.0e+00	7.0e+00	1.0e+01	1.4e+01	1.5e+01	5.1e-01
nearest_hydro_distance_m	6.2e+03	3.3e+03	1.1e+07	0	1.5e+03	3.7e+03	5.8e+03	8.4e+03	1.2e+04	2.1e+04	5.4e-01
nearest_hydro_river_stage_max_m	7.1e-01	1.3e+00	1.7e+00	-4.2e+00	-3.8e+00	3.9e-01	9.0e-01	1.4e+00	2.0e+00	2.5e+00	-2.6e+00
nearest_hydro_quota	1.1e+02	1.0e+02	1.1e+04	0	0	2.5e+01	1.0e+02	1.5e+02	3.2e+02	3.2e+02	7.1e-01
idw_temp_max_peak	2.6e+01	1.8e+00	3.4e+00	2.0e+01	2.3e+01	2.4e+01	2.6e+01	2.8e+01	2.8e+01	3.0e+01	-3.3e-01
idw_temp_min_nadir	2.2e+00	1.0e+00	1.1e+00	-2.0e-01	7.6e-01	1.1e+00	2.4e+00	3.2e+00	3.6e+00	4.2e+00	-1.2e-01
idw_temp_thermal_range	2.4e+01	1.1e+00	1.2e+00	1.9e+01	2.2e+01	2.3e+01	2.4e+01	2.5e+01	2.5e+01	2.7e+01	-2.6e-01
idw_rain_mm_sum_annual	6.4e+02	1.3e+02	1.6e+04	4.5e+02	4.9e+02	5.2e+02	6.0e+02	7.5e+02	8.5e+02	1.1e+03	4.8e-01
idw_rain_mm_max_monthly	8.1e+00	2.3e+00	5.2e+00	5.2e+00	5.7e+00	6.4e+00	7.2e+00	9.2e+00	1.3e+01	1.6e+01	1.2e+00
idw_rain_mm_min_monthly	1.1e-01	9.1e-02	8.2e-03	2.0e-04	2.1e-02	4.4e-02	7.2e-02	1.5e-01	3.0e-01	6.6e-01	1.7e+00
idw_rain_mm_avg_monthly	3.0e+00	6.0e-01	3.6e-01	2.1e+00	2.3e+00	2.5e+00	2.8e+00	3.5e+00	4.0e+00	5.0e+00	4.8e-01
idw_rain_mm_std	2.1e+00	4.8e-01	2.3e-01	1.4e+00	1.5e+00	1.7e+00	1.9e+00	2.4e+00	3.0e+00	3.5e+00	7.0e-01
idw_wind_speed_max_max	1.2e+01	3.7e+00	1.4e+01	7.0e+00	8.6e+00	9.3e+00	1.0e+01	1.6e+01	1.9e+01	3.0e+01	8.3e-01
idw_wind_speed_max_95p	1.1e+01	2.7e+00	7.2e+00	6.9e+00	8.0e+00	8.6e+00	9.4e+00	1.3e+01	1.5e+01	2.6e+01	1.2e+00
idw_wind_speed_avg_mean	2.2e+00	7.2e-01	5.2e-01	1.0e+00	1.4e+00	1.7e+00	1.9e+00	2.7e+00	3.5e+00	6.7e+00	1.2e+00
dtmidcnt_mean	3.7e+02	2.8e+02	7.6e+04	0	3.8e+01	1.5e+02	3.0e+02	5.5e+02	9.1e+02	1.6e+03	8.3e-01
dtmidcnt_max	3.9e+02	2.8e+02	8.0e+04	0	4.0e+01	1.7e+02	3.2e+02	5.8e+02	9.4e+02	1.7e+03	8.0e-01